

The Three-Dimensional Navier–Stokes Regularity Problem

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Abstract

We prove that for every fixed viscosity $\nu > 0$ and every divergence-free initial datum $u_0 \in \mathcal{S}(\mathbb{R}^3; \mathbb{R}^3)$, the maximal classical solution of the unforced three-dimensional incompressible Navier–Stokes equations is global, with velocity and pressure smooth on $\mathbb{R}^3 \times [0, \infty)$.

Write $[0, T_*)$ for the maximal classical lifespan and suppose that $T_* < \infty$. The central datum-generated estimate is uniform up to T_* : for every finite $N, M \in \mathbb{N}_0$,

$$\sum_{n=0}^N \left(\|\partial_t^n u\|_{L^2(0, T_*; H^{M+1})}^2 + \|\partial_t^{n+1} u\|_{L^2(0, T_*; H^{M-1})}^2 \right) < \infty.$$

At $(N, M) = (0, 1)$, the Hilbert-space trace theorem gives $u \in C([0, T_*]; H^1(\mathbb{R}^3))$. Sobolev embedding then gives $u \in L^\infty(0, T_*; L^3(\mathbb{R}^3))$. The Escauriaza–Seregin–Šverák endpoint criterion makes this bound incompatible with a finite maximal time.

Under the same finite-time hypothesis, the finite-order velocity and normalized-pressure families agree on overlaps, and their compatible intersection gives the stronger endpoint statement: $u(t) \rightarrow u_*$ in every $H^m(\mathbb{R}^3)$ as $t \uparrow T_*$, for one divergence-free $u_* \in \bigcap_{m < \infty} H^m(\mathbb{R}^3)$. At T_* , the differentiated equations recover every velocity derivative and the corresponding pressure derivatives under the fixed decaying normalization. Local well-posedness and uniqueness then restart the same solution from u_* . The L^3 contradiction and the smooth restart both rule out a finite T_* , so $T_* = \infty$. The global kinetic-energy identity gives $\|u(t)\|_2 \leq \|u_0\|_2$ for every $t \geq 0$.

Contents

1	The Regularity Problem	1
1.1	The equation and the target	1
2	Critical Height and the Derivative Tower	2
2.1	The critical-height problem inside a stretching vortex	3
2.2	Velocity, acceleration, jerk, and the Tower	6
2.3	The Tower opens toward more spatial regularity	7
3	The Intersection Inside the Tower	10
3.1	A spatial column inside every temporal response	10
3.2	The two-dimensional derivative grid	11
3.3	The finite piece that reaches a time slice	12
3.4	The estimate that must survive the endpoint	14
4	The Missing Estimate and Global Smoothness	16
4.1	The intersection reaches one terminal datum	16
4.2	The first rectangle seen from the endpoint	17
4.3	The critical-height roof	18
4.4	The two vortex readings	19
4.5	The singularity conditions fail at both scales	20
5	Further Consequences and the Euler Boundary	22
5.1	A finite sum across infinitely many scales	22
5.2	Why the Euler Tower is different	23
A	Local Construction and Classical Estimates	24
A.1	The local construction behind the maximal history	24
A.2	The Classical Estimates Before Closure	28
B	Derivative and Finite-Record Calculus	33
B.1	Critical Height at a Finite Endpoint	33
B.2	What finite time implies after one response escapes	34
B.3	The Pressure-Complete Mixed Jet	35
B.4	The Spatial Column Inside Critical Height	36
B.5	Simultaneous Sobolev Intersections	38
B.6	Finite Adjacent Rectangles	40
C	Compatible Terminal Assembly	43
C.1	The Terminal Quantifier	43
C.2	The Compatible Intersection	46
C.3	The Critical Height at the Terminal Face	47
C.4	One Endpoint for the Whole Tower	49
C.5	Restarting the Same History	50
	Bibliography	51

1 The Regularity Problem

The question begins with an ordinary promise and one extraordinary doubt. We are allowed to start the fluid as smoothly as we like. We are not allowed to assume that this smoothness survives the motion it creates. Before following that motion toward a stretching vortex, we fix exactly what was put into the equation, what the equation is being asked to return, and where the known energy control stops short of that return.

1.1 The equation and the target

Fefferman's formulation of the whole-space problem begins with an initial velocity that leaves no roughness to blame later (Fefferman 2000). It is smooth, divergence-free, and every one of its spatial derivatives dies faster than any polynomial as $|x|$ moves out through the fluid. Write this class as

$$\mathcal{S}_\sigma := \left\{ a \in C^\infty(\mathbb{R}^3; \mathbb{R}^3) : \begin{array}{l} \nabla \cdot a = 0, \\ \forall \alpha \in \mathbb{N}_0^3 \forall K \in \mathbb{N}_0 \exists C_{\alpha,K} < \infty \forall x \in \mathbb{R}^3, \\ |\partial_x^\alpha a(x)| \leq C_{\alpha,K} (1 + |x|)^{-K} \end{array} \right\}. \quad (1.1)$$

Read the condition in the order it is written. Choose any spatial derivative α . Choose any algebraic decay rate K , however large. The datum still has a finite constant $C_{\alpha,K}$ that controls that derivative everywhere in space. The constant changes when either choice changes; it does not change with x . This is the divergence-free Schwartz class.

For a finite spatial height m , let

$$H_\sigma^m(\mathbb{R}^3) := \{v \in H^m(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot v = 0\}.$$

The decay in (1.1) places the same datum in every one of these spaces. To see it, fix $K > 3/2$. For $|\alpha| \leq m$,

$$\|\partial_x^\alpha a\|_2^2 \leq C_{\alpha,K}^2 \int_{\mathbb{R}^3} (1 + |x|)^{-2K} dx < \infty.$$

Only finitely many derivatives occur below a chosen finite m , and hence

$$\mathcal{S}_\sigma \subset H_\sigma^m(\mathbb{R}^3) \quad \text{for every finite } m. \quad (1.2)$$

At time zero, then, every finite Sobolev height is already present. The regularity problem asks whether the evolution can destroy that complete smooth structure in finite time.

Fix one $u_0 \in \mathcal{S}_\sigma$ and one viscosity $\nu > 0$. The velocity u and the pressure p obey one coupled law:

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(\cdot, 0) = u_0. \quad (1.3)$$

The four parts should be kept together. The time derivative records the response of the field. Transport lets the velocity carry itself through the velocity it has already created. Incompressibility forces pressure to answer for that motion across the surrounding field.

Viscosity acts on the spatial curvature of the same velocity. There is one fluid here, and every later derivative row is another reading of this law.

Local theory starts that fluid smoothly. Uniqueness then joins the local pieces into one maximal classical history on an interval $[0, T_*)$, where $T_* \in (0, \infty]$. The Millennium Prize problem is the claim that this interval never has a finite right edge. In the notation above, its target is

$$T_* = \infty, \quad u, p \in C^\infty(\mathbb{R}^3 \times [0, \infty)), \quad \sup_{t \geq 0} \int_{\mathbb{R}^3} |u(x, t)|^2 dx < \infty. \quad (1.4)$$

Smoothness includes the last slice of every finite interval. A field can be smooth on every shorter interval $[0, T]$, $T < T_*$, while one of its derivatives grows without a finite trace as $t \uparrow T_*$. Interior smoothness and arrival at the endpoint are different statements. The proof has to carry the derivatives to that face or show that the face cannot exist.

There is one global estimate available before we do any of that. Pairing (1.3) with u makes transport and pressure disappear from the total kinetic-energy balance, while viscosity keeps its sign. The calculation in Proposition A.2.1 gives

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 ds = \frac{1}{2} \|u_0\|_2^2. \quad (1.5)$$

This controls the amount of velocity at each time and the accumulated gradient over time. It does not control the gradient at each time. A steeper and steeper change can live on shorter and shorter intervals while the integral in (1.5) remains finite.

That is the opening left by the classical estimate. The energy of the fluid can stay calm while its motion is being pressed across a narrowing distance. In three dimensions, vortex stretching is the part of the equation that can attempt exactly that. We follow that attempt next, from the geometry of the vortex to the scale on which energy stops seeing it, and then to the temporal responses the fluid would need in order to complete the concentration.

2 Critical Height and the Derivative Tower

Section 1 tells us what a global solution has to satisfy. It does not yet tell us what the fluid would have to do in order to lose that regularity. We begin with the three-dimensional mechanism capable of sharpening rotation while viscosity is diffusing it: vortex stretching. We follow that mechanism through its scale, the time available at that scale, and the responses the equation must produce if the sharpening is to continue. The derivative Tower is not imposed on this picture in advance. It appears when the proposed singularity is asked to account for its own motion.

2.1 The critical-height problem inside a stretching vortex

Take a region of fluid carrying coherent rotation and place it in an extensional strain. Write

$$\omega := \nabla \times u, \quad S := \frac{1}{2}(\nabla u + (\nabla u)^\top).$$

The curl of the momentum equation gives

$$\partial_t \omega + (u \cdot \nabla) \omega = S \omega + \nu \Delta \omega. \quad (2.1.1)$$

The term $S \omega$ is the exact action of strain on vorticity. When ω points along an extensional direction of S ,

$$\omega \cdot S \omega > 0,$$

and the rotation intensifies.

The vortex does not receive that amplification without changing its shape. Incompressibility says

$$\text{tr } S = \nabla \cdot u = 0.$$

Extension along the vortex axis must be accompanied by contraction in the transverse directions. The structure becomes longer, its cross-section becomes narrower, and the velocity variation carried across that cross-section is pressed into less space. If ΔU is the variation across a width ℓ , then

$$|\nabla u| \sim \frac{\Delta U}{\ell}.$$

The regularity problem is already visible in this ratio. The amount of velocity may stay finite while the distance over which it changes collapses.

Kinetic energy is allowed to miss that event. Its exact balance is

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 ds = \frac{1}{2} \|u_0\|_2^2. \quad (2.1.2)$$

The first term measures velocity over volume. It does not retain the width across which that velocity changes. A narrowing region can carry less and less kinetic energy while its gradients become steeper.

Viscosity cannot be discussed apart from that narrowing. Pair (2.1.1) with ω . The pointwise form is

$$\frac{1}{2}(\partial_t + u \cdot \nabla)|\omega|^2 = \omega \cdot S \omega + \frac{\nu}{2} \Delta |\omega|^2 - \nu |\nabla \omega|^2.$$

After integration over $\mathbb{R}^3$, transport and the Laplacian flux disappear, and the competition becomes

$$\frac{1}{2} \frac{d}{dt} \|\omega(t)\|_2^2 + \nu \|\nabla \omega(t)\|_2^2 = \int_{\mathbb{R}^3} \omega \cdot S \omega dx. \quad (2.1.3)$$

Stretching and diffusion are acting on the same rotating structure. As its width falls, both terms strengthen. The balance has shown us the mechanism, but it has not told us whether viscosity gains on the stretching.

Shrink the entire equation

Now shrink the picture without changing the law that governs it. For $\lambda > 0$, define

$$u_\lambda(x, t) := \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) := \lambda^2 p(\lambda x, \lambda^2 t). \quad (2.1.4)$$

Every term in the momentum equation receives the factor λ^3 :

$$\begin{aligned} \partial_t u_\lambda &= \lambda^3 (\partial_t u)(\lambda x, \lambda^2 t), & (u_\lambda \cdot \nabla) u_\lambda &= \lambda^3 ((u \cdot \nabla) u)(\lambda x, \lambda^2 t), \\ \nabla p_\lambda &= \lambda^3 (\nabla p)(\lambda x, \lambda^2 t), & \Delta u_\lambda &= \lambda^3 (\Delta u)(\lambda x, \lambda^2 t). \end{aligned}$$

Nothing in the smaller copy has handed viscosity an extra power of scale. Transport, pressure, and diffusion remain locked in the original proportion. The scale change has instead shortened the clock by λ^{-2} .

To see what the shrinking leaves unchanged, let $\Lambda = (-\Delta)^{1/2}$. A change of variables gives

$$\|u_\lambda(t)\|_{\dot{H}^s}^2 = \lambda^{2s-1} \|u(\lambda^2 t)\|_{\dot{H}^s}^2. \quad (2.1.5)$$

The exponent vanishes at one height:

$$2s - 1 = 0, \quad s = \frac{1}{2}.$$

We call

$$H(t) := \frac{1}{2} \|u(t)\|_{\dot{H}^{1/2}}^2 = \frac{1}{2} \|\Lambda^{1/2} u(t)\|_2^2 \quad (2.1.6)$$

the critical height. It is the spatial reading that a lawful concentration cannot make smaller by shrinking itself. Energy sits below it and loses one power of λ ; the first derivative sits above it and gains one:

$$\|u_\lambda(t)\|_2^2 = \lambda^{-1} \|u(\lambda^2 t)\|_2^2, \quad H_\lambda(t) = H(\lambda^2 t), \quad \|\nabla u_\lambda(t)\|_2^2 = \lambda \|\nabla u(\lambda^2 t)\|_2^2.$$

This is why finite energy can remain calm while a derivative sharpens. The critical height occupies the point between those two readings where the concentration is still fully visible.

One rescaled profile makes the separation concrete. For a nonzero divergence-free $\phi \in \mathcal{S}_\sigma$, set

$$u_\ell(x) := \ell^{-1} \phi(x/\ell).$$

Then

$$\begin{aligned} \|u_\ell\|_\infty &\sim \ell^{-1}, & \|\nabla u_\ell\|_\infty &\sim \ell^{-2}, \\ \|u_\ell\|_2^2 &\sim \ell, & \|u_\ell\|_{\dot{H}^{1/2}}^2 &= \|\phi\|_{\dot{H}^{1/2}}^2. \end{aligned}$$

The amplitude and gradient rise, the energy falls, and the critical height does not move. This family is not a Navier–Stokes trajectory. It shows the shape of the loophole left open by energy and respected by scaling.

What the equation does at critical height

Scaling has selected the right height. The equation now tells us how that height moves. For

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2,$$

keep transport and pressure together in the signed response

$$\mathcal{P}_s(t) := -\operatorname{Re} \left\langle \Lambda^s u, \Lambda^s \left((u \cdot \nabla) u + \nabla p \right) \right\rangle_{L^2}.$$

The pressure part vanishes in this global pairing after incompressibility is used. We keep it visible because the scalar balance came from the complete momentum equation, not from transport alone. Pairing at height s gives

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1}. \quad (2.1.7)$$

At the critical height,

$$H' = \mathcal{P}_{1/2} - 2\nu E_{3/2}. \quad (2.1.8)$$

Positive production raises H only after paying the viscous loss beside it. The equation has placed the unresolved stretching–diffusion contest at the scale-invariant height where neither side can escape by rescaling.

Give the shrinking scale a clock

The linear heat flow assigns an exact time scale to a width. If $v_t = \nu \Delta v$, then

$$\hat{v}(\xi, t + \delta t) = e^{-\nu |\xi|^2 \delta t} \hat{v}(\xi, t).$$

A structure of width r lives near $|\xi| \simeq r^{-1}$. Diffusion changes its amplitude by order one on the time

$$\tau_\nu(r) \asymp \frac{r^2}{\nu}. \quad (2.1.9)$$

This is not yet an estimate on the nonlinear fluid. It is the amount of time viscosity assigns to a scale once the evolution has actually formed that scale.

Take geometrically shrinking widths

$$r_n := 2^{-n} r_0, \quad \tau_n := \frac{r_n^2}{\nu}.$$

Their heat clocks satisfy

$$\tau_n = \frac{r_0^2}{\nu} 4^{-n}, \quad \sum_{n=0}^{\infty} \tau_n = \frac{4r_0^2}{3\nu} < \infty. \quad (2.1.10)$$

This is the Zeno opening: the clocks attached to infinitely many smaller scales fit inside a finite amount of elapsed time.

The arithmetic does not create a singularity. The fluid would have to form r_0 , then r_1 , then r_2 , continuing without losing the stretching alignment while viscosity is already acting on every scale that has appeared. Each new event would begin from the velocity produced by the previous one. The widths shrink, the clocks shorten, and the critical stretching–diffusion balance has to be reproduced again.

That is what turns the spatial picture into a question about response. If the velocity is to cross sharper states during shorter intervals, what must its acceleration do? If the acceleration is forced upward, what must lead it? The singularity has brought us to the temporal derivative Tower.

2.2 Velocity, acceleration, jerk, and the Tower

The Zeno picture compresses space and time together. A smaller width is not reached by naming a later scale; the velocity has to change enough to produce it during the shorter interval available. For any spatial norm X in which the response is classical,

$$\|u(t_1) - u(t_0)\|_X \leq \int_{t_0}^{t_1} \|\partial_t u(s)\|_X ds. \quad (2.2.1)$$

An unbounded sequence of velocity changes inside a finite accumulated time cannot be produced by finite accumulated acceleration. This is not a claim that acceleration has one sign. It is the causal statement that the change in velocity has to be carried by its time derivative.

Apply the same statement to acceleration. If acceleration is itself forced through unbounded changes, then its time derivative—jerk—must carry those changes. The question repeats:

$$u, \quad \partial_t u, \quad \partial_t^2 u, \quad \partial_t^3 u, \quad \dots$$

Write

$$V_n := \partial_t^n u, \quad Q_n := \partial_t^n p, \quad n \in \mathbb{N}_0. \quad (2.2.2)$$

These are the velocity, acceleration, jerk, and higher responses of the fluid. The Tower is not an extra model placed beside Navier–Stokes. The original equation generates every rung.

At the base,

$$V_1 = \nu \Delta V_0 - (V_0 \cdot \nabla) V_0 - \nabla Q_0.$$

Differentiate once:

$$V_2 = \nu \Delta V_1 - (V_1 \cdot \nabla) V_0 - (V_0 \cdot \nabla) V_1 - \nabla Q_1.$$

Differentiate again, and the two ways of placing one derivative on each transport factor produce the middle coefficient:

$$V_3 = \nu \Delta V_2 - (V_2 \cdot \nabla) V_0 - 2(V_1 \cdot \nabla) V_1 - (V_0 \cdot \nabla) V_2 - \nabla Q_2.$$

The general row is the binomial expansion of the transport term:

$$V_{n+1} + \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} + \nabla Q_n = \nu \Delta V_n, \quad \nabla \cdot V_n = 0. \quad (2.2.3)$$

Pressure does not create a second Tower. Taking the divergence of the base equation and using the decaying whole-space normalization gives

$$-\Delta p = \partial_i \partial_j (u_i u_j), \quad p = R_i R_j (u_i u_j), \quad (2.2.4)$$

where $R_j = \partial_j (-\Delta)^{-1/2}$. After n time derivatives,

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} (V_a)_i (V_{n-a})_j. \quad (2.2.5)$$

The velocity rows determine Q_n ; its gradient returns to (2.2.3) as part of the next response. Every rung retains transport, pressure, viscosity, and incompressibility.

Now look at what the recurrence asks us to carry. The next temporal response contains ΔV_n . Transport differentiates products of the earlier responses. Pressure is recovered from spatial derivatives of those products. Climbing in time has already opened a spatial direction inside every rung.

To record that direction, let

$$J_{n,\alpha} := \partial_t^n \partial_x^\alpha u, \quad \Pi_{n,\alpha} := \partial_t^n \partial_x^\alpha p, \quad \alpha \in \mathbb{N}_0^3. \quad (2.2.6)$$

The temporal and spatial product rules give

$$\begin{aligned} J_{n+1,\alpha} + \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta} + \nabla \Pi_{n,\alpha} \\ = \nu \Delta J_{n,\alpha}, \quad \nabla \cdot J_{n,\alpha} = 0. \end{aligned} \quad (2.2.7)$$

The matching pressure coordinate is fixed by

$$-\Delta \Pi_{n,\alpha} = \partial_x^\alpha \partial_i \partial_j \sum_{a=0}^n \binom{n}{a} (V_a)_i (V_{n-a})_j. \quad (2.2.8)$$

This is the first reversal in the argument. We entered the Tower because a singularity would require the fluid to respond faster and faster. But every new response is generated through spatial derivatives, and viscosity places two more spatial derivatives of the current response directly in the row. The route toward faster temporal change is pointing upward in spatial order. We now return to the critical height and ask exactly how far upward it points.

2.3 The Tower opens toward more spatial regularity

Here is the strange turn. We followed the vortex toward smaller scales because that is where spatial control was supposed to fail. The Newtonian question produced velocity, acceleration, jerk, and the higher temporal responses. Yet the equation generating those responses keeps moving through the Laplacian, and the Laplacian moves upward in spatial order.

The critical height lets us see the two directions on the same scale. Under (2.1.4),

$$H_\lambda(t) = H(\lambda^2 t), \quad H_\lambda^{(n)}(t) = \lambda^{2n} H^{(n)}(\lambda^2 t).$$

The spatial energy at height $n + \frac{1}{2}$ carries exactly that weight:

$$E_{n+1/2,\lambda}(t) = \lambda^{2n} E_{n+1/2}(\lambda^2 t).$$

Scaling has placed the n -th temporal response of H beside the $(n + \frac{1}{2})$ -spatial energy. The equation tells us why they belong together.

Start from the exact balance already found in (2.1.7):

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1}.$$

At critical height,

$$H' = \mathcal{P}_{1/2} - 2\nu E_{3/2}.$$

Differentiate this identity and use the balance at the next height:

$$\begin{aligned} H'' &= \partial_t \mathcal{P}_{1/2} - 2\nu E'_{3/2} \\ &= \partial_t \mathcal{P}_{1/2} - 2\nu \mathcal{P}_{3/2} + (2\nu)^2 E_{5/2}. \end{aligned}$$

One more derivative gives

$$H''' = \partial_t^2 \mathcal{P}_{1/2} - 2\nu \partial_t \mathcal{P}_{3/2} + (2\nu)^2 \mathcal{P}_{5/2} - (2\nu)^3 E_{7/2}.$$

Nothing below the new top term has disappeared. Every differentiated transport–pressure response remains in the row, while viscosity exposes one higher spatial energy. For $n \geq 1$, induction gives

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}. \quad (2.3.1)$$

Since $\nu > 0$, the top spatial energy can be read back from the completed row:

$$E_{n+1/2} = \frac{H^{(n)} - \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}}{(-2\nu)^n}. \quad (2.3.2)$$

This is an identity, not a free estimate. The transfer terms have not been discarded, and each term on the right still has to be controlled. What has changed is the direction of the problem: higher temporal response does not lead us toward lower spatial order. It exposes

$$E_{1/2}, \quad E_{3/2}, \quad E_{5/2}, \quad E_{7/2}, \quad \dots$$

inside the completed critical-height Tower.

The scalar height rows remain tied to the velocity responses. Because

$$H(t) = \frac{1}{2} \left\langle \Lambda^{1/2} V_0(t), \Lambda^{1/2} V_0(t) \right\rangle_{L^2},$$

Leibniz's rule gives

$$H^{(n)}(t) = \frac{1}{2} \sum_{a=0}^n \binom{n}{a} \operatorname{Re} \left\langle \Lambda^{1/2} V_a(t), \Lambda^{1/2} V_{n-a}(t) \right\rangle_{L^2}. \quad (2.3.3)$$

The n -th height derivative is not the velocity response V_n , and it is not the spatial energy $E_{n+1/2}$. It is the quadratic contraction through which the temporal responses expose that energy once every transport–pressure term is retained.

Now fix one response V_r . The same calculation begins again inside that row. Set

$$E_{r,s} := \frac{1}{2} \|\Lambda^s V_r\|_2^2$$

and let $\mathcal{P}_{r,s}$ be the complete pairing of V_r with the transport and pressure terms in its differentiated equation. Then

$$E'_{r,s} = \mathcal{P}_{r,s} - 2\nu E_{r,s+1}. \quad (2.3.4)$$

Repeated differentiation opens the spatial column

$$V_r \in H_x^0, \quad V_r \in H_x^1, \quad V_r \in H_x^2, \quad V_r \in H_x^3, \quad \dots$$

at every fixed temporal order r . Temporal order chooses the response; spatial order chooses how finely that response is resolved. They are two independent directions inside one mixed derivative structure.

This is where the intersection enters. If one fixed response belongs to every finite spatial Sobolev space, then

$$\bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

If every finite temporal response carries that spatial column, the resulting object is

$$\bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I; H^m(\mathbb{R}^3)). \quad (2.3.5)$$

For any prescribed temporal derivative and spatial derivative, one chooses a finite coordinate high enough to reach it and then applies the corresponding Sobolev embedding. Smoothness is not being assigned to a chosen row. It is the common content of the complete intersection.

The discovery is now visible. A singularity asks the fluid to ascend through faster temporal responses while losing spatial regularity. The viscous recurrence opens the opposite grading: every response contains a direction toward higher spatial regularity, and every spatial column contains the smoothness intersection.

Those computations identify what must be assembled; they have not yet paid for passage to the terminal time. We now formalize the two-dimensional grid, cut it into finite adjacent rectangles, and isolate the endpoint estimate.

3 The Intersection Inside the Tower

Section 2 did not discover one privileged Sobolev estimate. It discovered two directions inside the equation. Temporal order moves through the responses $V_0, V_1, V_2, \dots$; spatial order moves through the Sobolev column carried by each response. We now make that relation exact, identify the finite pieces that possess time traces, and assemble those pieces without changing their shared coordinates.

3.1 A spatial column inside every temporal response

We begin with the base response $V_0 = u$, not because it is privileged, but because its critical height is the place where the two directions were first visible. For $s \geq 0$, set

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2$$

and retain the complete transport–pressure pairing

$$\mathcal{P}_s(t) := -\operatorname{Re} \left\langle \Lambda^s u, \Lambda^s \left((u \cdot \nabla) u + \nabla p \right) \right\rangle_{L^2}.$$

The spatial energy row is

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1}. \quad (3.1.1)$$

At $s = \frac{1}{2}$,

$$H' = \mathcal{P}_{1/2} - 2\nu E_{3/2}. \quad (3.1.2)$$

The next two rows make the triangular structure explicit:

$$H'' = \partial_t \mathcal{P}_{1/2} - 2\nu \mathcal{P}_{3/2} + (2\nu)^2 E_{5/2}, \quad (3.1.3)$$

$$H''' = \partial_t^2 \mathcal{P}_{1/2} - 2\nu \partial_t \mathcal{P}_{3/2} + (2\nu)^2 \mathcal{P}_{5/2} - (2\nu)^3 E_{7/2}. \quad (3.1.4)$$

Every lower transfer survives while the top spatial energy rises by one.

The complete identity is

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}, \quad n \geq 1. \quad (3.1.5)$$

Its induction from (3.1.1) is proved in Proposition B.4.1. Solving the triangular row for its top coordinate gives

$$E_{n+1/2} = \frac{H^{(n)} - \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}}{(-2\nu)^n}. \quad (3.1.6)$$

The division is lawful because $\nu > 0$. It does not erase the transfer sum. The completed row names the exact collection that has to be bounded if the exposed spatial energy is to be bounded.

The height derivative remains a contraction of the velocity responses:

$$H^{(n)} = \frac{1}{2} \sum_{a=0}^n \binom{n}{a} \operatorname{Re} \left\langle \Lambda^{1/2} V_a, \Lambda^{1/2} V_{n-a} \right\rangle_{L^2}. \quad (3.1.7)$$

The three readings at order n are distinct. V_n is the Eulerian response, $H^{(n)}$ is its quadratic critical contraction, and $E_{n+1/2}$ is the highest spatial energy exposed after the full transfer chain has been retained.

Table 3.1.1: The temporal response and the spatial height exposed at the same order.

order	velocity response	critical response	exposed height
0	$V_0 = u$	$H = E_{1/2}$	$E_{1/2}$
1	$V_1 = u_t$	H'	$E_{3/2}$
2	$V_2 = u_{tt}$	H''	$E_{5/2}$
3	$V_3 = u_{ttt}$	H'''	$E_{7/2}$
n	$V_n = \partial_t^n u$	$H^{(n)}$	$E_{n+1/2}$

Now shift the base of the calculation from u to any fixed response V_r . The Laplacian still acts linearly on that response, and every split of the transport derivative remains in the binomial sum. The next subsection lays those independent temporal and spatial choices out as one grid.

3.2 The two-dimensional derivative grid

Fix $r \in \mathbb{N}_0$. The scalar energy of the response V_r at spatial height s is

$$E_{r,s} := \frac{1}{2} \|\Lambda^s V_r\|_2^2.$$

The transfer paired with that response is

$$\mathcal{P}_{r,s} := -\operatorname{Re} \left\langle \Lambda^s V_r, \Lambda^s \left[\sum_{a=0}^r \binom{r}{a} (V_a \cdot \nabla) V_{r-a} + \nabla Q_r \right] \right\rangle_{L^2}.$$

Pairing the r -th differentiated momentum equation at this height produces

$$E'_{r,s} = \mathcal{P}_{r,s} - 2\nu E_{r,s+1}. \quad (3.2.1)$$

Beginning at $s = \frac{1}{2}$ and repeating the induction proved in Proposition B.4.1 gives the completed row

$$\partial_t^n E_{r,1/2} = (-2\nu)^n E_{r,n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{r,j+1/2}. \quad (3.2.2)$$

There is a spatial ladder inside u , another inside u_t , another inside u_{tt} , and one inside every finite response. Moving down changes the temporal response. Moving across changes the spatial height.

Table 3.2.1: The mixed derivative grid. Each entry is a coordinate of u ; no entry is yet an endpoint estimate.

	H_x^0	H_x^1	H_x^2	H_x^3	$\dots$
$V_0 = u$	•	•	•	•	$\dots$
$V_1 = u_t$	•	•	•	•	$\dots$
$V_2 = u_{tt}$	•	•	•	•	$\dots$
$V_3 = u_{ttt}$	•	•	•	•	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

The point of the grid is not to add its rows into an infinite norm. Take one finite request: q time derivatives and k spatial derivatives. Choose finite indices

$$r > q + \frac{1}{2}, \quad m > k + \frac{3}{2}.$$

The vector-valued time embedding followed by the three-dimensional spatial embedding (Cerdà 2010, Theorem 7.28) reads

$$H^r(I; H^m(\mathbb{R}^3)) \hookrightarrow C^q(\bar{I}; C^k(\mathbb{R}^3)).$$

Every finite request has a finite coordinate above its thresholds. Collecting all of them on the fluid gives

$$\mathcal{V}_\infty(I) := \bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I; H^m(\mathbb{R}^3)) \hookrightarrow C^\infty(\bar{I} \times \mathbb{R}^3). \quad (3.2.3)$$

The constants may depend on the selected (r, m) . Smoothness asks that every finite coordinate exist; it does not ask for one constant uniform over the entire grid.

The remaining issue is time. A dot in the grid can be finite on every strict subinterval and still diverge as the endpoint is approached. To carry a spatial coordinate to one time slice, we have to place it beside the temporal response directly below it. Those adjacent pairs determine the finite rectangles.

3.3 The finite piece that reaches a time slice

Let $I_* = (0, T_*)$ denote the interval on a proposed finite maximal branch, and first stop at a strict cutoff $T < T_*$. A spatial row by itself has no reason to arrive continuously at $t = T$. It needs its time derivative on the other side of the desired trace space.

Let $A = (1 - \Delta)^{1/2}$. For a smooth Hilbert-valued function v ,

$$\|v(b)\|_{H^M}^2 - \|v(a)\|_{H^M}^2 = 2 \operatorname{Re} \int_a^b \langle A^{M-1} v_t(t), A^{M+1} v(t) \rangle_{L^2} dt. \quad (3.3.1)$$

Cauchy–Schwarz places the middle trace between the two adjacent rows

$$v \in L^2(0, T; H^{M+1}), \quad v_t \in L^2(0, T; H^{M-1}), \quad (3.3.2)$$

and the conclusion is

$$v \in C([0, T]; H^M); \quad (3.3.3)$$

the approximation argument and strong representative are proved in Theorem C.3.1.

At $M = 2$, the first concrete pair is

$$u \in L^2(0, T; H^3), \quad u_t \in L^2(0, T; H^1) \implies u \in C([0, T]; H^2). \quad (3.3.4)$$

One row sits above the desired trace and the next temporal response sits below it. Their pairing reaches the middle.

Now apply that trace to V_n , whose derivative is V_{n+1} . To retain the responses through order N at a common height M , collect

$$\begin{aligned} \mathfrak{R}_{N,M}(u; T) := & \sum_{n=0}^N \|V_n\|_{L^2(0,T;H^{M+1})}^2 \\ & + \sum_{n=0}^N \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2. \end{aligned} \quad (3.3.5)$$

When $M = 0$, the lower edge is read in H^{-1} . The temporal depth N and spatial height M remain independent choices.

The first nontrivial example, $(N, M) = (1, 2)$, contains two adjacent traces sharing V_1 :

Table 3.3.1: The $(1, 2)$ rectangle. The shared response closes the first trace and begins the second.

response	upper row	lower row
$V_0 = u$	$V_0 \in L^2 H^3$	$V_1 \in L^2 H^1$
$V_1 = u_t$	$V_1 \in L^2 H^3$	$V_2 \in L^2 H^1$

Write $\mathcal{R}_{N,M}[u_0; T]$ for this velocity block together with the pressure rows reconstructed by (2.2.5) and the differentiated equations that identify every entry. The rectangle is a finite piece of the derivative grid, not a new fluid object.

Classical smoothness on a strict subinterval makes each chosen rectangle finite:

$$\mathfrak{R}_{N,M}(u; T) < \infty \quad (T < T_*, N, M < \infty). \quad (3.3.6)$$

But the terminal question changes the order of the quantifiers:

$$\forall T < T_* : \quad \mathfrak{R}_{N,M}(u; T) < \infty \quad \not\Rightarrow \quad \sup_{0 < T < T_*} \mathfrak{R}_{N,M}(u; T) < \infty.$$

After (N, M) is fixed, the proof needs one bound that survives every cutoff approaching T_* . This is the whole-terminal rectangle estimate.

3.4 The estimate that must survive the endpoint

Theorem 3.4.1 (Datum-generated whole-terminal rectangles). *Let $u_0 \in \mathcal{S}_\sigma$, let $\nu > 0$, and let (u, p) be the maximal classical solution generated by (u_0, ν) on $[0, T_*)$. If $T_* < \infty$, then for every finite $N, M \in \mathbb{N}_0$,*

$$\sup_{0 < T < T_*} \mathfrak{R}_{N,M}(u; T) < \infty. \quad (3.4.1)$$

The associated pressure rows are generated by (2.2.5) and use the fixed decaying normalization.

Remark 3.4.2 (Reading the terminal quantifier). The pair (N, M) is fixed before the supremum over $T < T_*$ is taken. The bound may depend on (N, M) ; no constant uniform over all derivative orders is asserted. Every rectangle belongs to the same history, and larger rectangles return the identical derivatives on their overlaps.

Everything before this theorem identifies the rectangle on each strict subinterval. That is not yet the estimate in (3.4.1). Classical smoothness allows the norm to grow each time the cutoff moves closer to T_* . The producer proof has to begin with (u_0, ν) , keep (N, M) fixed, and derive one bound that does not move with the cutoff.

That derivation is the load-bearing payment of the paper. The proof source currently attached to this manuscript states the datum-generated theorem but does not display the calculation that produces even its first rectangle. The terminal argument in Proposition C.1.2 starts from (3.4.1); it realizes the assumed rows on I_* and cannot serve as their producer. Until the missing datum-to-rectangle calculation is inserted here, every conclusion below that uses Theorem 3.4.1 remains conditional on this one theorem.

Fix N and M . Every term in (3.3.5) is an integral of a nonnegative squared norm. For any one integrand f , the intervals $(0, T)$ exhaust I_* , so

$$\int_{I_*} f(t) dt = \lim_{T \uparrow T_*} \int_0^T f(t) dt.$$

The whole-terminal estimate keeps that limit finite for every row in the chosen rectangle:

$$V_n \in L^2(I_*; H^{M+1}), \quad V_{n+1} = \partial_t V_n \in L^2(I_*; H^{M-1}), \quad 0 \leq n \leq N. \quad (3.4.2)$$

The pressure part of Proposition C.1.2, using the Riesz bounds and finite-order Sobolev products, gives the matching rows

$$Q_n \in L^1(I_*; H^M), \quad 0 \leq n \leq N. \quad (3.4.3)$$

Together these velocity and pressure coordinates form $\mathcal{R}_{N,M}[u_0]$ on the whole interval I_* .

A larger rectangle adds coordinates without replacing the ones already present. If $N' \leq N$ and $M' \leq M$, let $\rho_{N',M'}^{N,M}$ forget the extra rows and lower the Sobolev height. Since every retained entry is the same derivative of u ,

$$\rho_{N',M'}^{N,M} \mathcal{R}_{N,M}[u_0] = \mathcal{R}_{N',M'}[u_0]. \quad (3.4.4)$$

This literal overlap lets the finite blocks be read as one compatible family:

$$\mathcal{I}[u_0] := \left(\mathcal{R}_{N,M}[u_0] \right)_{N,M \in \mathbb{N}_0}, \quad \rho_{N',M'}^{N,M} \mathcal{R}_{N,M}[u_0] = \mathcal{R}_{N',M'}[u_0]. \quad (3.4.5)$$

The intersection creates no new field. Given any finite temporal order and spatial height, choose one rectangle large enough to contain that coordinate; every larger rectangle returns the same coordinate of the same u .

Proposition 3.4.3 (The mixed intersection). *The velocity component of $\mathcal{I}[u_0]$ satisfies*

$$u \in \bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I_*; H^m(\mathbb{R}^3)). \quad (3.4.6)$$

In particular,

$$u \in \bigcap_{m=0}^{\infty} L^2(I_*; H^m(\mathbb{R}^3)). \quad (3.4.7)$$

Proof. Fix finite r, m , and choose $N \geq r$ and $M \geq m$. The upper edge of this one rectangle gives

$$\partial_t^n u \in L^2(I_*; H^{M+1}) \hookrightarrow L^2(I_*; H^m) \quad (0 \leq n \leq r).$$

These are the distributional time derivatives of the same velocity field, hence $u \in H^r(I_*; H^m)$. Because r and m were arbitrary, every finite coordinate belongs to the intersection; $r = 0$ gives (3.4.7). $\square$

4 The Missing Estimate and Global Smoothness

The Tower has reached the place where a smooth solution can still disappear. Every finite rectangle exists on every shorter interval, and every overlap contains the same derivatives of the same fluid. Neither fact carries those norms to T_* . If the cutoff-dependent bounds rise without limit, the intersection never reaches the terminal face and there is no datum from which to restart.

This is why the order of the quantifiers in (3.4.1) matters. We fix one finite rectangle first and then move the cutoff. Its bound may know u_0 , ν , T_* , N , and M . It cannot know how close T is to T_* . Once that estimate is paid, the compatible rectangles become one mixed intersection, the intersection reaches one smooth terminal datum, and the same Navier–Stokes history restarts from it. We follow that complete argument before returning to the critical height, the Zeno ladder, and the narrowing vortex that forced us to build the Tower.

4.1 The intersection reaches one terminal datum

Theorem 4.1.1 (Whole-space existence and smoothness). *For every $\nu > 0$ and every $u_0 \in \mathcal{S}_\sigma$, the maximal classical solution of (1.3) satisfies*

$$T_* = \infty, \quad u, p \in C^\infty(\mathbb{R}^3 \times [0, \infty)). \quad (4.1.1)$$

For every $t \geq 0$,

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 ds = \frac{1}{2} \|u_0\|_2^2. \quad (4.1.2)$$

Proof. Suppose the maximal history ends at some finite T_* . For each finite pair (N, M) , Theorem 3.4.1 carries its pressure-complete rectangle all the way across I_* . Whenever two rectangles overlap, (3.4.4) says that they contain the same derivatives of the same history. Their compatible union gives

$$u \in \bigcap_{r, m < \infty} H^r(I_*; H^m(\mathbb{R}^3)).$$

Now hold one spatial height m fixed. The temporal row $r = 1$ puts u in $H^1(I_*; H^m)$, so the actual history has an H^m limit as $t \uparrow T_*$. A higher spatial row cannot produce a different limit: after embedding it into H^m , both limits come from the same $u(t)$. All finite heights meet at one divergence-free endpoint

$$u_* := \lim_{t \uparrow T_*} u(t) \in H_\sigma^\infty(\mathbb{R}^3) \subset \bigcap_{m < \infty} H^m(\mathbb{R}^3). \quad (4.1.3)$$

This is the endpoint furnished by Lemma C.4.1.

The terminal face must also remember how the history arrives there. Given finite N and m , the row $r = N + 1$ places $V_N = \partial_t^N u$ in $H^1(I_*; H^m)$. It has a trace V_N^* , and spatial

compatibility puts that trace in H^∞ . The normalized pressure recurrence and the velocity recurrence survive the limit:

$$\begin{aligned} Q_N^* &= R_i R_j \sum_{a=0}^N \binom{N}{a} V_{a,i}^* V_{N-a,j}^*, \\ V_{N+1}^* &= \nu \Delta V_N^* - \sum_{a=0}^N \binom{N}{a} (V_a^* \cdot \nabla) V_{N-a}^* - \nabla Q_N^*. \end{aligned}$$

The traces are the velocity and Riesz-normalized pressure responses generated by u_* , as Proposition C.4.2 records.

Use u_* as initial data at time T_* . The smooth local construction in Lemma C.5.1 gives a right-hand history on $[T_*, T_* + \delta]$. Its recurrence begins from the same datum and the same pressure normalization, so its entire jet agrees with the left-hand traces. The join is smooth. Strong uniqueness makes it the continuation of the history generated by u_0 , contradicting the definition of T_* .

Hence $T_* = \infty$, and the normalized pair is smooth on every finite time interval. The kinetic-energy identity in Proposition A.2.1 now holds on every initial segment of this global history and gives (4.1.2). In particular,

$$\sup_{t \geq 0} \int_{\mathbb{R}^3} |u(x, t)|^2 dx \leq \int_{\mathbb{R}^3} |u_0(x)|^2 dx < \infty.$$

The rapid decay of u_0 pays the last finiteness. This is the smooth global history with uniformly bounded kinetic energy required in (1.4). $\square$

The theorem uses the whole intersection because that is the object the Tower produced. Its lowest rectangle gives a second, independent way to see why the alleged last time has nowhere to form.

4.2 The first rectangle seen from the endpoint

Choose the first rectangle, $(N, M) = (0, 1)$. Its adjacent rows are

$$u \in L^2(I_*; H^2), \quad u_t \in L^2(I_*; L^2).$$

Their trace lands in the space between them:

$$u \in C([0, T_*]; H^1).$$

Since $H^1(\mathbb{R}^3) \hookrightarrow L^3(\mathbb{R}^3)$, this gives

$$\sup_{0 \leq t \leq T_*} \|u(t)\|_3 < \infty. \tag{4.2.1}$$

But the endpoint theorem of Escauriaza, Seregin, and Šverák says that a finite maximal time forces the opposite behaviour (Escauriaza, G. A. Seregin, and Šverák 2003):

$$T_* < \infty \implies \sup_{0 < t < T_*} \|u(t)\|_3 = \infty. \tag{4.2.2}$$

The rectangle bounds the same velocity in the same norm on the same terminal interval. The two conclusions cannot coexist, so this first coordinate already contradicts a finite T_* .

The full intersection keeps every higher response available, but the physical return does not need an inventory of them. The first rectangle has already reached the critical velocity norm. The next one reaches the motion of the critical height and the gradient that stretches the vortex. Those are the two coordinates we now follow.

4.3 The critical-height roof

Once the H^1 trace reaches the terminal face, the critical height cannot keep climbing underneath it. Indeed, $H = \frac{1}{2} \|\Lambda^{1/2} u\|_2^2$, so

$$H_{\max} := \sup_{0 \leq t \leq T_*} H(t) \leq \frac{1}{2} \sup_{0 \leq t \leq T_*} \|u(t)\|_{H^1}^2 < \infty. \quad (4.3.1)$$

Start at any positive height H_0 , double it successively by $L_k := 2^k H_0$, and count the levels the same history actually reaches:

$$N_{\text{levels}} := \#\{k \in \mathbb{N}_0 : \text{there is a } t \in [0, T_*] \text{ with } H(t) = L_k\}.$$

The roof leaves room for only finitely many such doublings:

$$N_{\text{levels}} \leq \begin{cases} 0, & H_{\max} < H_0, \\ 1 + \lfloor \log_2(H_{\max}/H_0) \rfloor, & H_{\max} \geq H_0. \end{cases} \quad (4.3.2)$$

This conclusion never asks how quickly one level is crossed. The summable heat clocks from Section 2 cannot assemble an infinite ladder because the height itself runs out of levels. A width might still shrink while the height stays bounded, so the geometric picture has one more question to answer.

The next rectangle, $\mathcal{R}_{0,2}$, follows the motion below the roof rather than only its maximum:

$$u \in L^2(I_*; H^3), \quad u_t \in L^2(I_*; H^1), \quad u \in C([0, T_*]; H^2). \quad (4.3.3)$$

The two adjacent rows now contain both factors in the derivative

$$H'(t) = \operatorname{Re} \langle \Lambda^{1/2} u(t), \Lambda^{1/2} u_t(t) \rangle_{L^2},$$

and Cauchy–Schwarz pays its total variation:

$$\int_0^{T_*} |H'(t)| dt \leq \|\Lambda^{1/2} u\|_{L_t^2 L_x^2} \|\Lambda^{1/2} u_t\|_{L_t^2 L_x^2} < \infty. \quad (4.3.4)$$

The same rectangle has $E_{3/2} \in L^1(I_*)$. Hence the completed critical row

$$\mathcal{P}_{1/2} = H' + 2\nu E_{3/2}$$

puts the signed production in $L^1(I_*)$, and integration records exactly how much of it has accumulated:

$$\int_0^t \mathcal{P}_{1/2}(s) ds = H(t) - H(0) + 2\nu \int_0^t E_{3/2}(s) ds. \quad (4.3.5)$$

The two terms on the right are already finite, so

$$A_*(T_*) := \sup_{0 < t < T_*} \int_0^t \mathcal{P}_{1/2}(s) ds < \infty. \quad (4.3.6)$$

Thus the signed prefix is read off after the rectangle has been built; it is not an upstream estimate from which the rectangle was obtained. Sharp rises and falls remain possible, but their net signed accumulation cannot diverge before T_* .

For a vortex, the decisive entry in $\mathcal{R}_{0,2}$ is $u \in L^2(I_*; H^3)$. In three dimensions $H^3 \hookrightarrow W^{1,\infty}$, whence

$$\int_0^{T_*} \|\nabla u(t)\|_\infty dt \leq T_*^{1/2} \|u\|_{L^2(I_*; H^3)} < \infty.$$

Both vorticity and strain are parts of this gradient. Their accumulated action before T_* is finite in the precise sense

$$\int_0^{T_*} (\|\nabla u(t)\|_\infty + \|\omega(t)\|_\infty + \|S(t)\|_\infty) dt < \infty. \quad (4.3.7)$$

4.4 The two vortex readings

What does a narrowing core mean for the fluid inside it? There are two answers to separate. The first follows the same material particles. The second allows the visible concentration to replace its particles as it moves. Neither exhausts singular geometry, but each makes a different claim that the rectangle bounds can test.

Follow the fixed labels first. Let $X(a, t)$ solve

$$\partial_t X(a, t) = u(X(a, t), t), \quad X(a, 0) = a.$$

Two such trajectories satisfy

$$\frac{d}{dt} |X(a, t) - X(b, t)| \geq -\|\nabla u(t)\|_\infty |X(a, t) - X(b, t)|,$$

and the finite Lipschitz action integrates this differential inequality to

$$|X(a, t) - X(b, t)| \geq |a - b| \exp\left(-\int_0^t \|\nabla u(s)\|_\infty ds\right) > 0 \quad (4.4.1)$$

Because the same row also puts u in $L^1(I_*; L^\infty)$, every trajectory has a limit at T_* . Letting $t \uparrow T_*$ preserves the lower bound:

$$|X(a, T_*) - X(b, T_*)| \geq |a - b| \exp\left(-\int_0^{T_*} \|\nabla u(s)\|_\infty ds\right) > 0.$$

Distinct labels remain spatially distinct at the terminal time. If a drawn core collapses to one point, it cannot still contain exactly the material core with which it began.

Even perfect centring cannot rescue a nonzero circulation from this fixed-label picture. The finite coordinate $(r, m) = (0, 3)$ gives

$$L_* := \sup_{0 \leq t \leq T_*} \|\nabla u(t)\|_\infty < \infty. \quad (4.4.2)$$

Relative velocity at any chosen centre $x_c(t)$ obeys

$$|u(t, x) - u(t, x_c)| \leq L_* |x - x_c|.$$

Around a circular cross-section $C_r(t)$, the constant centre velocity integrates to zero. What remains is bounded by its Lipschitz variation:

$$\left| \oint_{C_r(t)} u(t, x) \cdot d\ell \right| \leq 2\pi L_* r^2 \longrightarrow 0 \quad (r \downarrow 0). \quad (4.4.3)$$

As r shrinks, the tangential velocity relative to the centre is $O(r)$ and the circuit has length $O(r)$. Their product is $O(r^2)$, so this centred material core cannot carry a nonzero limiting circulation.

Now let membership change. Particle separation no longer tracks the visible core, but repeated recruitment would have to keep transferring critical content to ever smaller wavelengths. That claim belongs to the high-frequency tail of the whole velocity field. For $s > 1/2$ and $K \geq 1$, define

$$H_{>K}(t) := \frac{1}{2} \int_{|\xi| > K} |\xi| |\hat{u}(t, \xi)|^2 d\xi.$$

For every $s > 1/2$, the tail satisfies

$$H_{>K}(t) \leq \frac{1}{2} K^{1-2s} \|u(t)\|_{\dot{H}^s}^2. \quad (4.4.4)$$

Choose one integer $m \geq s$. The compatible assembly and adjacent trace proved in Proposition 3.4.3 and theorem C.3.1, followed by $H^m \hookrightarrow \dot{H}^s$, make the estimate uniform in time and force

$$\lim_{K \rightarrow \infty} \sup_{0 \leq t \leq T_*} H_{>K}(t) = 0. \quad (4.4.5)$$

Recruitment may move activity between scales, but it cannot leave a fixed amount of critical height beyond every cutoff. The two readings fail for different reasons: fixed labels cannot collapse and their centred circulation vanishes; changing labels cannot sustain the required ultraviolet critical tail.

4.5 The singularity conditions fail at both scales

The first rectangle already reaches the global signature of a singular time. The endpoint theorem of Escauriaza, Seregin, and Šverák says that a finite maximal time forces the critical

L^3 norm to escape (Escauriaza, G. A. Seregin, and Šverák 2003), as recorded in (4.2.2). The same velocity has the opposite bound in (4.2.1). A finite terminal time cannot satisfy both statements.

The local signature disappears from the same intersection. Fix a finite $T > 0$. The mixed velocity bounds and the normalized pressure recurrence in Proposition C.4.2 give two finite coordinates with the common ceiling

$$K_T := \sup_{0 \leq t \leq T} (\|u(t)\|_{L^\infty} + \|p(t)\|_{L^\infty}) < \infty. \quad (4.5.1)$$

Now place a parabolic cylinder around any point of the history:

$$Q_\rho(z_0) = B_\rho(x_0) \times (t_0 - \rho^2, t_0) \subset \mathbb{R}^3 \times (0, T].$$

Its spatial volume and time length contribute ρ^5 . Dividing by the critical factor ρ^2 still leaves three powers of the radius:

$$\rho^{-2} \iint_{Q_\rho(z_0)} (|u|^3 + |p|^{3/2}) \, dx \, dt \leq C \rho^3 (K_T^3 + K_T^{3/2}) \longrightarrow 0 \quad (\rho \downarrow 0). \quad (4.5.2)$$

A Caffarelli–Kohn–Nirenberg singular point would require a positive scale-invariant amount of velocity-pressure content to survive along shrinking cylinders (Caffarelli, Kohn, and Nirenberg 1982). Here that content vanishes.

This is the return promised by the vortex picture. The alleged endpoint has become a smooth time slice. The critical height has a finite roof, so the Zeno ladder has only finitely many levels. The stretching action is integrable, a material core cannot collapse its labels, and the local velocity-pressure content loses the mass a singular point would need at every smaller scale.

5 Further Consequences and the Euler Boundary

The four-part argument is complete once the intersection has returned to the critical height, the Zeno ladder, and the singularity conditions. Two further projections remain useful. One adds the vorticity across all dyadic scales; the other removes viscosity and shows which rung of the Tower then disappears.

5.1 A finite sum across infinitely many scales

Fix a finite $T > 0$. On $[0, T]$, the next spatial row $\mathcal{R}_{0,2}[u_0; T]$ puts u in $L_t^2 H_x^3$. That one extra row is enough to add the vorticity from every dyadic wavelength rather than inspect the shells one at a time.

Let $(\Delta_j)_{j \geq -1}$ be an inhomogeneous Littlewood–Paley decomposition (Bahouri, Chemin, and Danchin 2011, Chapter 2). At high frequency, $\Delta_j \omega$ sees wavelengths of order 2^{-j} , while its L^∞ norm records the largest rotation in that shell. Define

$$\|\omega\|_{B_{\infty,1}^0} := \sum_{j \geq -1} \|\Delta_j \omega\|_\infty.$$

Bernstein’s inequality costs three halves of a derivative in passing from L^2 to L^∞ :

$$\|\Delta_j \omega\|_\infty \leq C 2^{3j/2} \|\Delta_j \omega\|_2.$$

The H^2 weight of ω supplies two derivatives. Splitting $2^{3j/2} = 2^{-j/2} 2^{2j}$ exposes the spare half derivative, and Cauchy–Schwarz sums it:

$$\begin{aligned} \sum_{j \geq 0} \|\Delta_j \omega\|_\infty &\leq C \left(\sum_{j \geq 0} 2^{-j} \right)^{1/2} \left(\sum_{j \geq 0} 2^{4j} \|\Delta_j \omega\|_2^2 \right)^{1/2} \\ &\leq C \|\omega\|_{H^2} \leq C \|u\|_{H^3}. \end{aligned}$$

The low shell is controlled directly by $\|\omega\|_2$. Thus one finite Sobolev coordinate pays the full ℓ^1 shell sum:

$$\|\omega\|_{B_{\infty,1}^0} \leq C \|u\|_{H^3}. \quad (5.1.1)$$

Cauchy–Schwarz in time now turns the $L_t^2 H_x^3$ row into

$$\sum_{j \geq -1} \int_0^T \|\Delta_j \omega(t)\|_\infty dt = \int_0^T \|\omega(t)\|_{B_{\infty,1}^0} dt < \infty. \quad (5.1.2)$$

Infinitely many shells may still become active. What cannot happen before this finite T is a nonsummable succession of their time-integrated peaks. The conclusion consumes only $\mathcal{R}_{0,2}$: it asserts neither an analytic radius nor a weighted estimate over all derivative orders.

5.2 Why the Euler Tower is different

Remove viscosity while keeping the two histories separate. Both equations can still be differentiated in time, so both possess a mixed jet. Write

$$V_n^{\text{NS}} := \partial_t^n u^{\text{NS}}, \quad Q_n^{\text{NS}} := \partial_t^n p^{\text{NS}},$$

and define $V_n^{\text{E}}, Q_n^{\text{E}}$ from an Euler solution with its own history. The two recurrences differ in one place:

$$V_{n+1}^{\text{NS}} + \sum_{a=0}^n \binom{n}{a} (V_a^{\text{NS}} \cdot \nabla) V_{n-a}^{\text{NS}} + \nabla Q_n^{\text{NS}} = \nu \Delta V_n^{\text{NS}}, \quad (5.2.1)$$

$$V_{n+1}^{\text{E}} + \sum_{a=0}^n \binom{n}{a} (V_a^{\text{E}} \cdot \nabla) V_{n-a}^{\text{E}} + \nabla Q_n^{\text{E}} = 0. \quad (5.2.2)$$

Transport and the nonlocal pressure response remain in both equations. Only the Navier–Stokes row contains $\nu \Delta V_n^{\text{NS}}$, which connects a temporal response to two more spatial derivatives of that same response.

The energy rows expose the same difference. Set

$$E_s^{\text{NS}} := \frac{1}{2} \|\Lambda^s u^{\text{NS}}\|_2^2, \quad E_s^{\text{E}} := \frac{1}{2} \|\Lambda^s u^{\text{E}}\|_2^2,$$

with the corresponding nonlinear transfer terms. Then

$$(E_s^{\text{NS}})' = \mathcal{P}_s^{\text{NS}} - 2\nu E_{s+1}^{\text{NS}}, \quad (E_s^{\text{E}})' = \mathcal{P}_s^{\text{E}}. \quad (5.2.3)$$

Repeated time differentiation of the Navier–Stokes row, followed by division by $(-2\nu)^n$, isolates $E_{n+1/2}^{\text{NS}}$ in (3.1.6). The Euler row still mixes time and space nonlinearly, but no signed coercive term singles out the next spatial height.

If an Euler history were independently known to lie in the analogous all-orders intersection, then every fixed Sobolev norm would be controlled and in particular it would satisfy the Beale–Kato–Majda condition $\int_0^T \|\omega^{\text{E}}(t)\|_\infty dt < \infty$ (Beale, Kato, and Majda 1984). The present theorem supplies no such Euler intersection, and none of its constants is asserted to remain bounded as $\nu \downarrow 0$.

That is where the argument ends. Euler keeps the jet, transport, pressure, and their nonlinear coupling; it loses the viscous rung that converts temporal response into a separately controllable spatial ascent. At every fixed $\nu > 0$, that ascent exposes the finite rectangles whose terminal bounds rule out a last time. The resulting Navier–Stokes history has every finite mixed coordinate on each finite interval, while its kinetic energy remains uniformly bounded for all time.

A Local Construction and Classical Estimates

A maximal history is not merely a chain of short solutions; its last time must be tied to a norm that the equation can test. The local construction makes that link, and the classical estimates show why finite kinetic energy alone cannot keep the history smooth.

A.1 The local construction behind the maximal history

One local theorem must do two jobs without changing its clock. It constructs the initial history and later restarts it from the terminal datum. The clock is set by ν and the datum's H^3 size; any higher regularity already carried by that datum survives on the same interval.

For real s , let

$$H_\sigma^s(\mathbb{R}^3) := \{v \in H^s(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot v = 0\}.$$

Proposition A.1.1 (Local construction and continuation alternative). *Fix $t_0 \geq 0$, $\nu > 0$, and $a \in H_\sigma^3(\mathbb{R}^3)$. A time $\delta = \delta(\nu, \|a\|_{H^3}) > 0$ exists for which there is a unique normalized-pressure pair (w, q) satisfying*

$$w \in C([t_0, t_0 + \delta]; H_\sigma^3) \cap L^2(t_0, t_0 + \delta; H^4), \quad w_t \in L^2(t_0, t_0 + \delta; H^2).$$

If the datum also lies in H_σ^m for an integer $m > 3$, this same interval carries

$$w \in C([t_0, t_0 + \delta]; H_\sigma^m) \cap L^2(t_0, t_0 + \delta; H^{m+1}), \quad w_t \in L^2(t_0, t_0 + \delta; H^{m-1}).$$

These local pairs agree on overlaps and form one maximal history. If its maximal time is finite, its H^3 norm is unbounded along times approaching that endpoint.

Proof of Proposition A.1.1. Keep t_0, ν, a fixed as above and measure all real Sobolev orders with

$$\|f\|_{H^s}^2 := \int_{\mathbb{R}^3} \langle \xi \rangle^{2s} |\widehat{f}(\xi)|^2 d\xi, \quad \langle \xi \rangle = (1 + |\xi|^2)^{1/2}.$$

Write $\mathbb{P}$ for the Leray projection and R_j for the Riesz transform with symbol $i\xi_j/|\xi|$. Each acts boundedly on every H^s .

The three rows forced by the nonlinearity and the heat equation

The construction must first absorb one derivative of $w \otimes w$. In three dimensions,

$$H^3(\mathbb{R}^3) \hookrightarrow W^{1,\infty}(\mathbb{R}^3), \quad \|fg\|_{H^3} \leq C \|f\|_{H^3} \|g\|_{H^3},$$

so an H^3 velocity is Lipschitz and products stay in H^3 . After the divergence and Leray projection this gives

$$\|\mathbb{P}\nabla \cdot (f \otimes g)\|_{H^2} \leq C \|f\|_{H^3} \|g\|_{H^3}. \quad (\text{A.1.1})$$

Leibniz' rule proves the estimate for integer derivatives, using $H^2 \hookrightarrow L^\infty$ and $H^1 \hookrightarrow L^6$; density then removes the smoothness assumption.

Once the force lies in H^2 , the heat equation places one parabolic derivative above the base state and the time derivative one below it. Thus $z \in L_t^2 H_x^4$ and $z_t \in L_t^2 H_x^2$, with H^3 exactly between them. For any such v ,

$$v \in L^2(J; H^4), \quad v_t \in L^2(J; H^2),$$

frequency truncation followed by the fundamental theorem of calculus and Cauchy–Schwarz yields

$$\left| \|v(t)\|_{H^3}^2 - \|v(r)\|_{H^3}^2 \right| \leq 2 \|v_t\|_{L^2(r,t;H^2)} \|v\|_{L^2(r,t;H^4)}. \quad (\text{A.1.2})$$

The right-hand side vanishes with the length of the time interval. Removing the frequency cutoff therefore gives a unique representative in $C(J; H^3)$, so the three rows of the strong class are

$$C_t H_x^3 \cap L_t^2 H_x^4, \quad \partial_t v \in L_t^2 H_x^2.$$

The continuous middle trace is what lets the equation remember its initial value while the heat operator gains one square-integrable derivative.

A linear estimate aligned with those rows

On $J_\delta = [t_0, t_0 + \delta]$, $0 < \delta \leq 1$, consider

$$z_t - \nu \Delta z = F, \quad z(t_0) = a, \quad F \in L^2(J_\delta; H^2).$$

At the H^3 level the force and solution occupy the adjacent spaces H^2 and H^4 . Their Fourier pairing is

$$\langle F, z \rangle_{H^2, H^4; H^3} := \int_{\mathbb{R}^3} \langle \xi \rangle^6 \widehat{F}(\xi) \cdot \overline{\widehat{z}(\xi)} d\xi.$$

The weight factors as $\langle \xi \rangle^6 = \langle \xi \rangle^2 \langle \xi \rangle^4$ and Cauchy–Schwarz therefore gives

$$|\langle F, z \rangle_{H^2, H^4; H^3}| \leq \|F\|_{H^2} \|z\|_{H^4}. \quad (\text{A.1.3})$$

For a smooth solution the corresponding energy row is

$$\frac{1}{2} \frac{d}{dt} \|z\|_{H^3}^2 + \nu \|\nabla z\|_{H^3}^2 = \text{Re} \langle F, z \rangle_{H^2, H^4; H^3}.$$

The inhomogeneous norm separates the dissipative part exactly:

$$\|z\|_{H^4}^2 = \|z\|_{H^3}^2 + \|\nabla z\|_{H^3}^2.$$

Insert (A.1.3), absorb by Young's inequality, and apply Gronwall to obtain

$$\|z\|_{C(J_\delta; H^3)} + \|z\|_{L^2(J_\delta; H^4)} \leq C_\nu \left(\|a\|_{H^3} + \|F\|_{L^2(J_\delta; H^2)} \right).$$

The equation then controls the missing time row as well:

$$\|z\|_{C(J_\delta; H^3)} + \|z\|_{L^2(J_\delta; H^4)} + \|z_t\|_{L^2(J_\delta; H^2)} \leq C_\nu \left(\|a\|_{H^3} + \|F\|_{L^2(J_\delta; H^2)} \right). \quad (\text{A.1.4})$$

No constant here deteriorates as δ shrinks within $(0, 1]$. Frequency-truncate a and F , solve the smooth equations, and apply (A.1.4) to differences. The Cauchy limit solves the untruncated problem, while (A.1.2) recovers its prescribed strong H^3 value at t_0 . Applying the estimate to the difference of two solutions proves uniqueness.

Closing the nonlinear map

Set

$$X_\delta := C(J_\delta; H_\sigma^3) \cap L^2(J_\delta; H^4), \quad \|w\|_{X_\delta} := \|w\|_{CH^3} + \|w\|_{L^2 H^4}.$$

The projected equation defines the map

$$\Phi(w)(t) := e^{\nu(t-t_0)\Delta} a - \int_{t_0}^t e^{\nu(t-s)\Delta} \mathbb{P} \nabla \cdot (w \otimes w)(s) ds. \quad (\text{A.1.5})$$

All abbreviated time norms below refer to J_δ . The product estimate and the factor $\delta^{1/2}$ from the interval give

$$\|\mathbb{P} \nabla \cdot (w \otimes w)\|_{L^2 H^2} \leq C \delta^{1/2} \|w\|_{CH^3}^2, \quad (\text{A.1.6})$$

$$\|\mathbb{P} \nabla \cdot (w \otimes w - v \otimes v)\|_{L^2 H^2} \leq C \delta^{1/2} (\|w\|_{CH^3} + \|v\|_{CH^3}) \|w - v\|_{CH^3}. \quad (\text{A.1.7})$$

The linear estimate turns these force bounds into

$$\|\Phi(w)\|_{X_\delta} \leq C_\nu \|a\|_{H^3} + C_\nu C \delta^{1/2} \|w\|_{X_\delta}^2$$

and

$$\|\Phi(w) - \Phi(v)\|_{X_\delta} \leq C_\nu C \delta^{1/2} (\|w\|_{X_\delta} + \|v\|_{X_\delta}) \|w - v\|_{X_\delta}.$$

When $a \neq 0$, set $R = 2C_\nu \|a\|_{H^3}$ and choose $\delta \leq 1$ with $4C_\nu C \delta^{1/2} R \leq 1$. Then Φ maps the closed radius- R ball into itself and contracts it by at least a factor of two. For $a = 0$, the fixed point is identically zero. Banach's theorem gives $w \in X_\delta$, with a choice satisfying

$$\delta \geq \min \left\{ 1, c_\nu (1 + \|a\|_{H^3})^{-2} \right\}. \quad (\text{A.1.8})$$

The constant $c_\nu > 0$ depends only on the fixed viscosity and the chosen Sobolev normalization.

Uniqueness is not confined to that ball. If $w, v \in X_\delta$ share the datum, apply the linear and difference estimates to $w - v$ on an initial subinterval short enough to absorb the coefficient. The solutions agree there. Their common endpoint starts the same argument on the next subinterval, and a finite partition covers J_δ . The heat semigroup and $\mathbb{P}$ preserve divergence-free fields; the projected equation also supplies

$$w_t \in L^2(J_\delta; H^2). \quad (\text{A.1.9})$$

Restoring the pressure to the same solution

For the constructed velocity, set

$$q := R_i R_j (w_i w_j). \quad (\text{A.1.10})$$

The algebra property and Riesz bounds imply

$$q \in C(J_\delta; H^3).$$

For $F = \nabla \cdot (w \otimes w)$, the Fourier symbols identify $\mathbb{P}F = F + \nabla q$. Hence the projected fixed point is the pressure-complete solution

$$w_t + (w \cdot \nabla)w + \nabla q = \nu \Delta w, \quad \nabla \cdot w = 0, \quad w(t_0) = a.$$

Membership of $q(t)$ in $H^3(\mathbb{R}^3)$ removes the whole-space additive constant. Two normalized pressures for the same velocity differ by an L^2 function with zero gradient and therefore coincide. The uniqueness already proved is consequently uniqueness of the pair (w, q) .

Higher regularity without a shorter clock

Fix an integer $m \geq 3$ and temporarily take a smooth. Differentiate the projected equation by every ∂^α , $|\alpha| \leq m$, pair with the same derivative of w , and sum. Divergence-free transport cancels at the top order:

$$\left((w \cdot \nabla) \partial^\alpha w, \partial^\alpha w \right)_2 = 0.$$

Every remaining Leibniz term differentiates the transporting factor at least once. The embeddings $H^2 \hookrightarrow L^\infty$ and $H^1 \hookrightarrow L^6$ then give

$$\left(\sum_{|\alpha| \leq m} \|[\partial^\alpha, w \cdot \nabla] w\|_2^2 \right)^{1/2} \leq C_m \|\nabla w\|_\infty \|w\|_{H^m}.$$

Thus the differentiated energy obeys

$$\frac{1}{2} \frac{d}{dt} \|w\|_{H^m}^2 + \nu \|\nabla w\|_{H^m}^2 \leq C_m \|\nabla w\|_\infty \|w\|_{H^m}^2. \quad (\text{A.1.11})$$

The already constructed $C_t H_x^3$ solution has integrable Lipschitz norm on this same interval. Gronwall therefore yields

$$\|w(t)\|_{H^m}^2 \leq \|a\|_{H^m}^2 \exp \left(2C_m \int_{t_0}^t \|\nabla w(s)\|_\infty ds \right). \quad (\text{A.1.12})$$

Retaining the dissipative term gives $w \in L^2(J_\delta; H^{m+1})$. At order $m-1$, the nonlinear term lies in $L^2(J_\delta; H^{m-1})$, so the equation also gives

$$w_t \in L^2(J_\delta; H^{m-1}).$$

The adjacent trace at height m completes the three rows:

$$w \in C(J_\delta; H^m) \cap L^2(J_\delta; H^{m+1}), \quad w_t \in L^2(J_\delta; H^{m-1}). \quad (\text{A.1.13})$$

The pressure formula and the H^m algebra property put q in $C(J_\delta; H^m)$ as well.

For general $a \in H_\sigma^m$, approximate it by divergence-free frequency cutoffs a_k . Their H^3 norms share a bound controlled by $\|a\|_{H^3}$; hence (A.1.8) puts every smooth approximation on one common interval. The fixed-point estimate identifies the H^3 limit with w , while the persistence inequalities bound all three higher rows uniformly. Weak compactness then identifies their limits with the same w . The adjacent trace gives continuity in H^m ; the energy inequality at t_0 , together with weak convergence to a , upgrades the initial convergence to strong convergence in H^m .

The interval never depends on m ; it was chosen once from ν and $\|a\|_{H^3}$. If $a \in H_\sigma^\infty$, repeat the argument at each finite height. Uniqueness in H^3 keeps every height on the same pair, and the pressure formula and momentum equation then recover every finite time derivative. Spatial embedding makes (w, q) smooth throughout the local space–time slab.

Joining the local pieces and locating the only obstruction

Begin at t_0 and extend through every interval allowed by the construction. Strong uniqueness makes the velocity and normalized pressure identical on overlaps, producing one pressure-complete history on a maximal interval $[t_0, T_{\max})$.

Suppose $T_{\max} < \infty$ and

$$\sup_{t_0 \leq t < T_{\max}} \|w(t)\|_{H^3} \leq M.$$

If its displayed H^3 bound were finite, the lifespan estimate would supply the same positive amount of future time

$$\delta_M := \min \left\{ 1, c_\nu (1 + M)^{-2} \right\}$$

from every later datum $w(t)$. Choose $t > T_{\max} - \delta_M/2$. Restarting at $w(t)$ then crosses $T_{\max}$, and uniqueness glues that restart to the supposed maximal history. The contradiction proves

$$T_{\max} < \infty \implies \limsup_{t \uparrow T_{\max}} \|w(t)\|_{H^3} = \infty. \quad (\text{A.1.14})$$

Every higher order already present in the datum persists along each piece, because every piece uses this same H^3 -controlled clock. In particular, $a \in \mathcal{S}_\sigma \subset H_\sigma^\infty$ generates the smooth normalized maximal history claimed above. $\square$

A.2 The Classical Estimates Before Closure

The local theorem has launched a smooth fluid. To keep it alive, we must know which part of that smoothness survives for as long as the solution exists. Kinetic energy survives globally, but at a derivative level too low to restart the classical history. The classical theory measures the exact distance between those two facts.

Energy reaches the endpoint before the gradient does

Proposition A.2.1 (Classical kinetic-energy identity). *Let (u, p) be a sufficiently decaying smooth solution of (1.3). At every time in its interval of existence,*

$$\|u(t)\|_{L^2}^2 + 2\nu \int_0^t \|\nabla u(s)\|_{L^2}^2 ds = \|u_0\|_{L^2}^2. \quad (\text{A.2.1})$$

Proof. Pair the momentum equation with u in space. The transport term carries energy through the fluid but cannot change its total amount:

$$\int_{\mathbb{R}^3} (u \cdot \nabla) u \cdot u \, dx = \frac{1}{2} \int_{\mathbb{R}^3} u \cdot \nabla |u|^2 \, dx = 0.$$

Pressure also does no total work, since

$$\int_{\mathbb{R}^3} \nabla p \cdot u \, dx = - \int_{\mathbb{R}^3} p \nabla \cdot u \, dx = 0.$$

The viscous term alone has a sign, and integration by parts gives

$$\nu \int_{\mathbb{R}^3} \Delta u \cdot u \, dx = -\nu \|\nabla u\|_{L^2}^2.$$

Time integration is exactly (A.2.1). □

Thus the velocity has a pointwise-in-time L^2 ceiling, while the gradient is paid only after its square has been accumulated over time.

Leray's construction preserves precisely this information without assuming the solution remains smooth. Regularization and compactness produce a global weak solution with (Leray 1934)

$$u \in L^\infty(0, T; L^2(\mathbb{R}^3)) \cap L^2(0, T; H^1(\mathbb{R}^3))$$

on each finite interval. The weak history is global, but this estimate by itself neither makes arbitrary three-dimensional Leray solutions unique nor returns the missing smoothness.

The spatial half of the energy class improves through

$$H^1(\mathbb{R}^3) \hookrightarrow L^6(\mathbb{R}^3)$$

and therefore

$$u \in L^2(0, T; L^6(\mathbb{R}^3)).$$

The nonlinear term consumes that gain immediately:

$$\|(u \cdot \nabla) u\|_{L^{3/2}} \leq \|u\|_{L^6} \|\nabla u\|_{L^2},$$

so transport lies in $L_t^1 L_x^{3/2}$, enough for the weak equation to make sense. It does not bound $\|\nabla u(t)\|_2$ at each time. Taller peaks can occupy shorter intervals and still leave the squared time integral finite.

In three dimensions, that missing instantaneous gradient is exactly where a vortex can amplify itself. Put

$$\omega := \nabla \times u.$$

For the divergence-free decaying fields at hand,

$$\|\nabla u\|_{L^2}^2 = \|\omega\|_{L^2}^2,$$

so the time-integrated gradient payment is the time-integrated enstrophy. Taking curl reveals its evolution:

$$\partial_t \omega + (u \cdot \nabla) \omega = (\omega \cdot \nabla) u + \nu \Delta \omega.$$

The same velocity gradient that the energy estimate controls only in time now stretches and tilts ω . Pairing this equation with ω gives

$$\frac{1}{2} \frac{d}{dt} \|\omega\|_{L^2}^2 + \nu \|\nabla \omega\|_{L^2}^2 = \int_{\mathbb{R}^3} (\omega \cdot \nabla) u \cdot \omega \, dx.$$

Nothing fixes the sign of the stretching integral, and kinetic energy does not force it below dissipation at each instant. The global estimate has therefore reached the whole interval without reaching the derivative height required for a classical restart.

The derivative height that buys another interval

For an integer $m \geq 0$, collect the first m spatial derivative levels in

$$\|u\|_{H^m}^2 := \sum_{|\alpha| \leq m} \|\partial^\alpha u\|_{L^2}^2,$$

and use the Fourier definition at fractional order. In three dimensions, Sobolev embedding gives (Cerdà 2010, Theorem 7.28)

$$H^s(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3) \quad \text{when} \quad s > k + \frac{3}{2}.$$

In particular,

$$s > \frac{5}{2} \quad \implies \quad H^s(\mathbb{R}^3) \hookrightarrow W^{1,\infty}(\mathbb{R}^3).$$

Above 5/2, one Sobolev norm controls the pointwise gradient that multiplies every differentiated energy row.

Suppose, then, that on $[0, T)$

$$\sup_{0 \leq t < T} \|u(t)\|_{H^s} \leq M \quad \text{for some } s > \frac{5}{2}.$$

At any fixed finite height $m \geq s$, differentiate the equation, pair each row with the matching derivative of u , and sum. Incompressibility cancels the top transport term, leaving

$$\frac{1}{2} \frac{d}{dt} \|u\|_{H^m}^2 + \nu \|\nabla u\|_{H^m}^2 \leq C_m \|\nabla u\|_{L^\infty} \|u\|_{H^m}^2.$$

The assumed H^s roof bounds the coefficient, so Gronwall carries this chosen H^m norm to time T . Its constant may depend on m : smoothness requires each finite height, not one uniform constant for infinitely many heights.

Once the velocity derivatives are present, pressure is forced by them. Taking divergence gives

$$-\Delta p = \sum_{i,j=1}^3 \partial_i \partial_j (u_i u_j),$$

with spatial decay fixing its additive constant. The elliptic relation returns the matching spatial derivatives of p ; the momentum equation then returns

$$\partial_t u = \nu \Delta u - (u \cdot \nabla) u - \nabla p$$

Repeating pressure recovery and the momentum equation produces every fixed mixed derivative. One time-uniform Sobolev roof above $5/2$ therefore recovers the full smooth velocity–pressure pair.

That roof also provides time. Local well-posedness in H^s has a lifespan controlled by s , ν , and the initial H^s size (Fujita and Kato 1964; Kato 1984). Restart at a late slice $u(\cdot, t_0)$. If all such slices remain below M , their lifespans share a positive lower bound; choosing t_0 close enough to T crosses it. Uniqueness joins the restart to the original history. Consequently, a finite maximal time can occur only if

$$\limsup_{t \uparrow T_*} \|u(t)\|_{H^s} = \infty \quad \text{for every } s > \frac{5}{2}.$$

The unpaid distance is now explicit: the energy class gives time-integrated H^1 , while continuation needs time-uniform control above $H^{5/2}$.

The scale on which concentration can hide

The Navier–Stokes scaling explains why energy cannot simply be upgraded to the missing roof. From any solution (u, p) , form

$$u_\lambda(x, t) := \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) := \lambda^2 p(\lambda x, \lambda^2 t)$$

which solves the same equation after space and time are contracted together. Any invariant norm sees the rescaled structure at the same size even while its physical width tends to zero.

The Ladyzhenskaya–Prodi–Serrin line balances the strength of that spatial concentration against the time for which it persists (Prodi 1959; Serrin 1962; Serrin 1963; Ladyzhenskaya 1967). A solution continues through T if

$$u \in L^q(0, T; L^p(\mathbb{R}^3)), \quad \frac{2}{q} + \frac{3}{p} \leq 1, \quad p > 3.$$

At equality, spatial intensity and temporal duration change by inverse factors under the Navier–Stokes scaling; the norm is critical.

The limiting spatial member is $L_t^\infty L_x^3$, since

$$\|u_\lambda(t)\|_{L^3} = \|u(\lambda^2 t)\|_{L^3}.$$

Escauriaza, Seregin, and Šverák showed that even this endpoint condition

$$u \in L^\infty(0, T; L^3(\mathbb{R}^3))$$

continues the solution through T (Escauriaza, G. A. Seregin, and Šverák 2003). Their argument rescales a first singularity into an ancient limit and eliminates it by backward uniqueness. Hence a finite last time must make the critical L^3 norm escape.

The same threat has a local description. Around $z_0 = (x_0, t_0)$, use the parabolic cylinder

$$Q_r(z_0) := B_r(x_0) \times (t_0 - r^2, t_0).$$

Caffarelli, Kohn, and Nirenberg proved that small scale-invariant velocity and pressure content on $Q_r(z_0)$ makes a smaller cylinder regular (Caffarelli, Kohn, and Nirenberg 1982). A singular point must therefore retain a definite amount of that content at every sufficiently small scale. Their covering argument makes the possible singular set small in parabolic Hausdorff measure, but one surviving point is enough to destroy global smoothness.

For smooth compactly supported whole-space data, Seregin sharpens the endpoint statement to actual divergence along the terminal approach (G. Seregin 2012):

$$\lim_{t \uparrow T_*} \|u(t)\|_{L^3(\mathbb{R}^3)} = \infty.$$

Tao's quantitative refinement forces a positive power of a triple logarithm of the inverse time remaining along a sequence of terminal times (Tao 2021). This lower rate is far slower than any power law; the classical results still permit much faster concentration.

The one estimate still absent

If the maximal time were finite, kinetic energy would remain bounded while the critical L^3 norm and every continuation-grade Sobolev norm escaped. Locally, some parabolic cylinders would retain nonzero scale-invariant content at arbitrarily small radii. These results describe what a singularity must do; they do not give the terminal-uniform estimate that prevents it.

The mismatch with energy is visible in the two scaling laws

$$\|u_\lambda(t)\|_{L^3} = \|u(\lambda^2 t)\|_{L^3}, \quad \|u_\lambda(t)\|_{L^2}^2 = \lambda^{-1} \|u(\lambda^2 t)\|_{L^2}^2.$$

A shorter structure retains its critical L^3 size while carrying less L^2 energy. A global kinetic-energy ceiling therefore cannot become a scale-independent ceiling on the critical concentration.

The same scaling shortens the clock. Repeatedly halving the active width gives parabolic durations proportional to

$$1, \frac{1}{4}, \frac{1}{16}, \dots,$$

whose sum is finite. Each stage can retain critical size while occupying less time, and none of the estimates above forbids that arithmetic. This is all the Zeno schedule asserts: it exposes an available timing, not a constructed singular Navier–Stokes history.

Closure therefore asks for one terminal-uniform continuation-grade norm on the actual maximal history.

B Derivative and Finite-Record Calculus

Strict classicality gives every derivative before T_* , but it does not tell us which finite part of that information can survive uniformly as the cutoff approaches the endpoint. The calculus here turns the derivative family into finite records so the terminal estimate has a precise object to keep bounded.

B.1 Critical Height at a Finite Endpoint

Return to the one maximal solution generated by the rapidly decaying datum u_0 at fixed $\nu > 0$, and suppose its last time T_* is finite. The field is smooth before that time. Its threat must therefore be a change of scale along the same history, not an already singular initial state.

Kinetic energy remains bounded, yet a structure’s contribution to that energy changes when its width changes. For $\lambda > 0$, set

$$u_\lambda(x, t) := \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) := \lambda^2 p(\lambda x, \lambda^2 t)$$

These fields solve Navier–Stokes with the same viscosity on the rescaled time interval. Space contracts by λ^{-1} , time by λ^{-2} , and velocity grows by λ ; one time derivative and the viscous two-space- derivative term consequently transform together.

Let $\Lambda = (-\Delta)^{1/2}$ and measure height by

$$\|v\|_{\dot{H}^s}^2 := \int_{\mathbb{R}^3} |\xi|^{2s} |\widehat{v}(\xi)|^2 d\xi.$$

Proposition B.1.1 (Critical Sobolev scaling). *Whenever the homogeneous norm is finite, every $s \in \mathbb{R}$ and $\lambda > 0$ satisfy*

$$\|u_\lambda(t)\|_{\dot{H}^s}^2 = \lambda^{2s-1} \|u(\lambda^2 t)\|_{\dot{H}^s}^2.$$

The only scale-invariant member is $s = \frac{1}{2}$.

Proof. The rescaled Fourier transform is

$$\widehat{u}_\lambda(\xi, t) = \lambda^{-2} \widehat{u}(\xi/\lambda, \lambda^2 t).$$

Substitution into the norm, followed by $\eta = \xi/\lambda$, gives

$$\begin{aligned}\|u_\lambda(t)\|_{\dot{H}^s}^2 &= \int_{\mathbb{R}^3} |\xi|^{2s} \lambda^{-4} \left| \widehat{u}(\xi/\lambda, \lambda^2 t) \right|^2 d\xi \\ &= \lambda^{2s-1} \int_{\mathbb{R}^3} |\eta|^{2s} \left| \widehat{u}(\eta, \lambda^2 t) \right|^2 d\eta.\end{aligned}$$

The prefactor is one exactly when $2s - 1 = 0$, proving the claim. $\square$

Compression therefore tilts every Sobolev height except $1/2$. Name the quantity that remains unchanged:

$$H(t) := \frac{1}{2} \|u(t)\|_{\dot{H}^{1/2}}^2 = \frac{1}{2} \left\| \Lambda^{1/2} u(t) \right\|_2^2.$$

An exact rescaling can now narrow and accelerate a smooth configuration without changing H .

At energy height, the same narrowing has the opposite weight:

$$\|u_\lambda(t)\|_2^2 = \lambda^{-1} \left\| u(\lambda^2 t) \right\|_2^2.$$

A finer structure can therefore retain its critical height while carrying less kinetic energy. The energy ceiling does not itself prevent the active width from continuing to fall.

The parabolic durations in Section A.2 allow successively finer scales to fit arithmetically before a finite endpoint. At critical height, those rescaled copies can all have the same H while their energy falls. They are still only copies: scaling has not made the actual solution visit them. Such a passage would have to be generated by transport, pressure, incompressibility, and viscosity acting together on the preceding state.

The scale is now fixed. A finite endpoint turns the remaining issue into response: if one row of the actual field escapes, what must the next time row do before the available time runs out?

B.2 What finite time implies after one response escapes

The equation supplies the temporal rows; finite time supplies the implication between two adjacent rows. If one response has no finite roof while the next is integrable, the fundamental theorem of calculus would nevertheless give the first a terminal limit. The following argument records that contradiction. It propagates escape through rows already known to exist and creates none of them.

Use one fixed Banach space B throughout and write the Eulerian rows as

$$V_n(t) := \partial_t^n u(\cdot, t),$$

Assume $V_n : [0, T_*) \rightarrow B$ is locally absolutely continuous and has Bochner derivative V_{n+1} . For $0 \leq s < t < T_*$,

$$V_n(t) - V_n(s) = \int_s^t V_{n+1}(\tau) d\tau$$

and therefore

$$\|V_n(t) - V_n(s)\|_B \leq \int_s^t \|V_{n+1}(\tau)\|_B d\tau.$$

Suppose

$$\sup_{0 < t < T_*} \|V_n(t)\|_B = \infty,$$

Since $\|V_n(0)\|_B$ is finite, the integral on the right cannot remain finite; hence

$$\int_0^{T_*} \|V_{n+1}(\tau)\|_B d\tau = \infty.$$

The interval itself is finite, so an L^1 divergence forces

$$\sup_{0 < t < T_*} \|V_{n+1}(t)\|_B = \infty.$$

Iteration is licensed only when every successive pair has this same fixed space, local absolute continuity, and Bochner-derivative identity. Under those hypotheses, finite time turns escape of one temporal response into escape of the next.

B.3 The Pressure-Complete Mixed Jet

At temporal order n , the equation keeps four pieces together: V_{n+1} , $\nu \Delta V_n$, every temporal split of transport, and the normalized pressure row Q_n . A single spatial derivative already shows how this row widens:

$$\partial_\ell \left((V_a \cdot \nabla) V_{n-a} \right) = ((\partial_\ell V_a) \cdot \nabla) V_{n-a} + (V_a \cdot \nabla) \partial_\ell V_{n-a}.$$

At multi-index order α , every derivative must be distributed between the transporting and transported factors. Denote the resulting coordinates by

$$J_{n,\alpha} := \partial_t^n \partial_x^\alpha u, \quad \Pi_{n,\alpha} := \partial_t^n \partial_x^\alpha p,$$

and use

$$\binom{\alpha}{\beta} := \prod_{k=1}^3 \binom{\alpha_k}{\beta_k}.$$

The two Leibniz rules then give the complete transport split

$$\partial_t^n \partial_x^\alpha \left((u \cdot \nabla) u \right) = \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta}.$$

Substitution keeps that split beside viscosity and pressure:

$$\begin{aligned} J_{n+1,\alpha} + \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta} \\ + \nabla \Pi_{n,\alpha} = \nu \Delta J_{n,\alpha}, \quad \nabla \cdot J_{n,\alpha} = 0. \end{aligned}$$

Taking divergence supplies the corresponding pressure row:

$$-\Delta \Pi_{n,\alpha} = \partial_i \partial_j \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta})_i (J_{n-a,\alpha-\beta})_j.$$

Every displayed coordinate is a derivative of the original pressure-complete pair. A finite rectangle will retain only finitely many of them, but it never changes the underlying solution.

The viscous coupling links the next temporal row to two additional spatial derivatives of the present row. The critical-height energy calculation isolates that ascent while preserving the transport and pressure terms that travel with it.

B.4 The Spatial Column Inside Critical Height

To follow the viscous link from a temporal response into space, take the scalar energy at an arbitrary spatial height $s \geq 0$:

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2,$$

In particular,

$$H(t) = E_{1/2}(t).$$

Keeping s free lets the same calculation be reused after the critical row.

Apply Λ^s to the full momentum equation and pair with $\Lambda^s u$. The temporal contribution is

$$\langle \Lambda^s u, \Lambda^s \partial_t u \rangle = \frac{d}{dt} \frac{1}{2} \|\Lambda^s u\|_2^2 = E'_s.$$

Viscosity commutes with Λ^s , and spatial decay gives

$$\nu \langle \Lambda^s u, \Lambda^s \Delta u \rangle = -\nu \|\Lambda^{s+1} u\|_2^2 = -2\nu E_{s+1}.$$

Pressure cancels in this global scalar balance only after integration by parts uses incompressibility:

$$\langle \Lambda^s u, \Lambda^s \nabla p \rangle = -\langle \nabla \cdot \Lambda^s u, \Lambda^s p \rangle = 0.$$

It remains present in the local equation and in every mixed coordinate. Accordingly, keep the signed transfer in its pressure-complete form:

$$\mathcal{P}_s(t) := -\left\langle \Lambda^s u, \Lambda^s \left((u \cdot \nabla) u + \nabla p \right) \right\rangle.$$

The resulting identity is

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1}.$$

One derivative in time has reached one full Sobolev level higher in space, but only together with the signed transfer at the original height.

At $s = 1/2$, this becomes

$$H' = \mathcal{P}_{1/2} - 2\nu E_{3/2}.$$

The scale-invariant height now touches $E_{3/2}$. Its mere appearance is not a bound: transfer and dissipation can cancel inside H' . Differentiate the whole row, not the isolated dissipative term.

The next energy row is

$$E'_{3/2} = \mathcal{P}_{3/2} - 2\nu E_{5/2},$$

and substitution into the derivative of the critical row yields

$$\begin{aligned} H'' &= \mathcal{P}'_{1/2} - 2\nu E'_{3/2} \\ &= \mathcal{P}'_{1/2} - 2\nu \mathcal{P}_{3/2} + (2\nu)^2 E_{5/2}. \end{aligned}$$

The response H'' reaches $E_{5/2}$, while retaining the transfer from each preceding height. One more derivative uses

$$E'_{5/2} = \mathcal{P}_{5/2} - 2\nu E_{7/2}$$

and gives

$$H''' = \mathcal{P}''_{1/2} - 2\nu \mathcal{P}'_{3/2} + (2\nu)^2 \mathcal{P}_{5/2} - (2\nu)^3 E_{7/2}.$$

The fourth response continues the same recurrence:

$$H^{(4)} = \mathcal{P}'''_{1/2} - 2\nu \mathcal{P}''_{3/2} + (2\nu)^2 \mathcal{P}'_{5/2} - (2\nu)^3 \mathcal{P}_{7/2} + (2\nu)^4 E_{9/2}.$$

At each step, the highest spatial energy rises once and the entire lower transfer history remains in the row.

This produces a triangular identity.

Proposition B.4.1 (Critical-height spatial ascent). *For every integer $n \geq 1$,*

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}.$$

Proof. The critical balance is the case $n = 1$. If the formula holds at order n , differentiate it and use

$$E'_{n+1/2} = \mathcal{P}_{n+1/2} - 2\nu E_{n+3/2}.$$

The new transfer extends the sum by one term, while the dissipative part becomes $(-2\nu)^{n+1} E_{n+3/2}$. This is the formula at order $n + 1$. $\square$

The proposition is an identity, not an estimate. Knowing $H^{(n)}$ cannot isolate $E_{n+1/2}$ until the complete signed transfer sum is controlled.

Critical scaling was homogeneous; continuation uses inhomogeneous Sobolev spaces. For $s > 0$, the two are joined by

$$\|u(t)\|_{H^s}^2 \simeq \|u(t)\|_2^2 + \|\Lambda^s u(t)\|_2^2 = \|u(t)\|_2^2 + 2E_s(t).$$

The kinetic-energy identity pays the low-frequency term:

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(\tau)\|_2^2 d\tau = \frac{1}{2} \|u_0\|_2^2.$$

Hence a bound for $E_s(t)$ completes an H^s bound, while time-integrability of E_s completes the high-frequency part of $L_t^2 H_x^s$ on a finite interval.

The ascent begins at $1/2$ because scaling selected it, and the first response reaches $3/2$. At the terminal face the logic reverses: whole-terminal Sobolev rows first bound these two heights, from which H and its first dissipative level follow.

Each finite number of responses reaches only a finite spatial height. To keep every finite coordinate of the derivative family, the same velocity must occupy every finite temporal and spatial Sobolev row.

B.5 Simultaneous Sobolev Intersections

The ascent has no final spatial height, although each concrete derivative request uses only finitely many rungs. To make all $\partial_x^\alpha u$ with $|\alpha| \leq k$ continuous, use the three-dimensional embedding

$$H^m(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3)$$

whenever

$$m > k + \frac{3}{2}.$$

Thus each requested order k chooses one finite height above its threshold. There is no largest m as k ranges over all finite orders, but no single infinite-order norm is needed. The proper object is the coordinatewise intersection

$$\bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3),$$

whose coordinates are all measured on the same field. For each fixed k , one coordinate supplies C^k , hence

$$\bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

Time behaves the same way. Let $I = (0, T)$, $T < T_*$, and let X be a Hilbert space.

The vector-valued embedding

$$H^r(I; X) \hookrightarrow C^q(\bar{I}; X)$$

holds for

$$r > q + \frac{1}{2}.$$

Each requested time derivative q chooses one finite r , and all finite requests give

$$\bigcap_{r=0}^{\infty} H^r(I; X) \hookrightarrow C^\infty(\bar{I}; X).$$

No common bound across all r is asserted.

The mixed jet makes the temporal and spatial choices simultaneously. For example, request

$$\partial_t^2 \partial_x^\alpha u, \quad |\alpha| \leq 4.$$

The one membership

$$u \in H^3(I; H^6(\mathbb{R}^3))$$

places this entire finite request in

$$C^2(\bar{I}; C^4(\mathbb{R}^3)),$$

since

$$3 > 2 + \frac{1}{2}, \quad 6 > 4 + \frac{3}{2}.$$

The pair $(2, 4)$ is only an example. For arbitrary finite q, k , choose finite indices with

$$r > q + \frac{1}{2}, \quad m > k + \frac{3}{2}.$$

Then

$$H^r(I; H^m(\mathbb{R}^3)) \hookrightarrow C^q(\bar{I}; C^k(\mathbb{R}^3)).$$

Collect every finite time-space request on the same velocity by defining

$$\mathcal{V}_\infty(I) := \bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I; H^m(\mathbb{R}^3)).$$

Explicitly,

$$u \in \mathcal{V}_\infty(I) \iff \|u\|_{H^r(I; H^m)} < \infty \text{ for every finite pair } (r, m).$$

The norms remain separate coordinates; the intersection does not sum them or replace u by a family of fields. The finite embeddings imply

$$\mathcal{V}_\infty(I) \hookrightarrow C^\infty(\bar{I} \times \mathbb{R}^3).$$

Choosing one temporal height above the trace threshold and one spatial height above $3/2$ places both critical levels on the closed interval:

$$u(t) \in H^{1/2}(\mathbb{R}^3) \quad \text{and} \quad u(t) \in H^{3/2}(\mathbb{R}^3)$$

Their scalar quantities are consequently defined there by

$$H(t) = \frac{1}{2} \left\| \Lambda^{1/2} u(t) \right\|_2^2, \quad E_{3/2}(t) = \frac{1}{2} \left\| \Lambda^{3/2} u(t) \right\|_2^2$$

Scaling and the first dissipative ascent selected these two heights; the intersection is what carries them to the terminal face.

Pressure is recovered alongside this velocity space, not inserted into it. At finite temporal depth write

$$Q_n = \partial_t^n p,$$

The Poisson equation recovers Q_n from the velocity rows, while the differentiated momentum equation keeps ∇Q_n beside response, viscosity, and transport. This produces a pressure-complete jet without misidentifying pressure as a velocity coordinate.

An actual estimate will control only a finite block. The two adjacent Sobolev levels required for a common time trace determine that block's upper and lower edges.

B.6 Finite Adjacent Rectangles

Only finite estimates can build the intersection. Begin on $I = (0, T)$, $T < T_*$, with the first request that needs adjacent rows: bring both

$$V_0 = u \quad \text{and} \quad V_1 = \partial_t u$$

to one time slice in $H^2(\mathbb{R}^3)$. The upper spatial row $V_0 \in L_t^2 H_x^3$ alone has no time trace. Its lower adjacent row is

$$\partial_t V_0 = V_1 \in L^2(I; H^1).$$

To trace V_1 at that same height, repeat the pair once:

$$V_1 \in L^2(I; H^3), \quad \partial_t V_1 = V_2 \in L^2(I; H^1).$$

Together these are

$$V_0, V_1 \in L^2(I; H^3), \quad V_1, V_2 \in L^2(I; H^1).$$

The repeated V_1 plays two roles: it closes the lower side of the first trace and occupies the upper side of the second.

The reason the levels straddle H^2 is visible directly. Let

$$A := (1 - \Delta)^{1/2}, \quad \|v\|_{H^s} = \|A^s v\|_2.$$

For smooth v ,

$$\|v(t)\|_{H^2}^2 - \|v(s)\|_{H^2}^2 = 2 \operatorname{Re} \int_s^t \left\langle A \partial_\tau v(\tau), A^3 v(\tau) \right\rangle_{L^2} d\tau.$$

The integrand is bounded by

$$\left| \left\langle A \partial_\tau v, A^3 v \right\rangle_{L^2} \right| \leq \|\partial_\tau v\|_{H^1} \|v\|_{H^3},$$

which is integrable in time whenever

$$v \in L^2(I; H^3), \quad \partial_t v \in L^2(I; H^1).$$

Approximation and weak continuity consequently give

$$v \in C([0, T]; H^2(\mathbb{R}^3)).$$

Applying this once to each adjacent pair yields

$$V_0, V_1 \in C([0, T]; H^2(\mathbb{R}^3)).$$

Collect the four norms consumed by these two traces in

$$\begin{aligned} \mathfrak{R}_{1,2}(u; T) := & \|u\|_{L^2(0,T;H^3)}^2 + \|\partial_t u\|_{L^2(0,T;H^3)}^2 \\ & + \|\partial_t u\|_{L^2(0,T;H^1)}^2 + \|\partial_t^2 u\|_{L^2(0,T;H^1)}^2. \end{aligned}$$

All four entries belong to the original solution. Their grouping is a record of what the two traces consume, not a construction of four new fields.

Now choose independent finite cutoffs $N, M \in \mathbb{N}_0$. To trace each retained temporal row at H^M , use the adjacent scale

$$H^{M+1}(\mathbb{R}^3) \hookrightarrow H^M(\mathbb{R}^3) \hookrightarrow H^{M-1}(\mathbb{R}^3),$$

with H^{-1} as the lower member when $M = 0$. For every $0 \leq n \leq N$, the required pair is

$$V_n \in L^2(0, T; H^{M+1}), \quad V_{n+1} = \partial_t V_n \in L^2(0, T; H^{M-1}).$$

Their squared finite velocity norm is

$$\begin{aligned} \mathfrak{R}_{N,M}(u; T) := & \sum_{n=0}^N \|V_n\|_{L^2(0,T;H^{M+1})}^2 \\ & + \sum_{n=0}^N \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2. \end{aligned}$$

For a smooth representative, each row satisfies

$$\left| \frac{d}{dt} \|V_n(t)\|_{H^M}^2 \right| \leq 2 \|V_{n+1}(t)\|_{H^{M-1}} \|V_n(t)\|_{H^{M+1}}.$$

Its right-hand side is integrable, so density and weak continuity yield

$$V_n \in C([0, T]; H^M(\mathbb{R}^3)), \quad 0 \leq n \leq N.$$

The temporal cutoff N chooses how many responses are retained; the spatial cutoff M chooses their common trace height. They remain independent, and the rectangle constant may depend on either one.

The scalar $\mathfrak{R}_{N,M}(u; T)$ remembers only the velocity norms. A rectangle must also remember why those rows belong together: retain the normalized pressure coordinates and the differentiated equations. For the pair generated by u_0 , define

$$\mathcal{R}_{N,M}[u_0; T] := \left((V_n)_{n=0}^{N+1}, (Q_n)_{n=0}^N; \mathfrak{R}_{N,M}(u; T) \right),$$

with

$$V_n = \partial_t^n u, \quad Q_n = \partial_t^n p.$$

Each retained temporal row obeys

$$V_{n+1} + \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} + \nabla Q_n = \nu \Delta V_n,$$

$$\nabla \cdot V_n = 0,$$

and

$$-\Delta Q_n = \partial_i \partial_j \sum_{a=0}^n \binom{n}{a} (V_a)_i (V_{n-a})_j.$$

Rapid decay fixes every pressure constant. At the retained depths, the mixed velocity coordinates are the weak derivatives

$$J_{n,\alpha} = \partial_x^\alpha V_n,$$

and the pressure coordinates are the corresponding derivatives of Q_n . Thus $\mathfrak{R}_{N,M}$ measures the finite velocity block, whereas $\mathcal{R}_{N,M}$ retains its pressure-complete equation identity.

To restrict such a block in time, let $J = (0, \tau)$, $0 < \tau \leq T_*$, and let $\mathcal{X}_{N,M}(J)$ consist of records with

$$W_n \in L^2(J; H^{M+1}), \quad W_{n+1} \in L^2(J; H^{M-1}) \quad (0 \leq n \leq N),$$

with $\partial_t W_n = W_{n+1}$ distributionally on $J \times \mathbb{R}^3$, and normalized pressure rows

$$P_n = R_i R_j \sum_{a=0}^n \binom{n}{a} (W_a)_i (W_{n-a})_j \in L^1(J; H^M) \quad (0 \leq n \leq N),$$

Their gradients lie in $L^1(J; H^{M-1})$, and the displayed momentum and divergence equations hold distributionally. The scalar component is the adjacent-row norm of the retained velocities on J . In particular,

$$\mathcal{R}_{N,M}[u_0; T] \in \mathcal{X}_{N,M}((0, T)).$$

As (N, M) ranges over finite pairs, the rectangles cover every finite coordinate of the intersection, each with its own finite bound. Overlapping rectangles contain the same distribution wherever they name the same derivative. That literal agreement is the datum needed for terminal assembly.

C Compatible Terminal Assembly

A bound for every finite record is still not a terminal solution. Because the records are obtained separately, their shared coordinates must remain the same as the rectangle and cutoff change; otherwise their limits need not belong to one history.

C.1 The Terminal Quantifier

Fix again the rapid-decay datum u_0 , the viscosity $\nu > 0$, and their single maximal normalized history (u, p) . Suppose its maximal time is finite and write $I_* = (0, T_*)$. Every coordinate below is a derivative of this history. The datum belongs to the smooth solenoidal class

$$H_\sigma^\infty(\mathbb{R}^3) := \left\{ v \in \bigcap_{m \in \mathbb{N}_0} H^m(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot v = 0 \right\}. \quad (\text{C.1.1})$$

Spatial decay fixes pressure from the velocity without an additive constant. With $R_j = \partial_j(-\Delta)^{-1/2}$,

$$-\Delta p = \partial_i \partial_j (u_i u_j), \quad p = R_i R_j (u_i u_j). \quad (\text{C.1.2})$$

Direct time differentiation, with $V_n = \partial_t^n u$ and $Q_n = \partial_t^n p$, gives

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}, \quad (\text{C.1.3})$$

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n. \quad (\text{C.1.4})$$

Every division of the time derivatives between the two transport factors remains in the binomial sum. Thus each temporal row still contains the full transport, pressure, viscosity, and divergence-free evolution.

Choose finite temporal depth N , spatial height M , and a strict cutoff $T < T_*$. The adjacent-row norm is

$$\begin{aligned} \mathfrak{R}_{N,M}(u; T) := & \sum_{n=0}^N \left\| \partial_t^n u \right\|_{L^2(0,T; H^{M+1})}^2 \\ & + \sum_{n=0}^N \left\| \partial_t^{n+1} u \right\|_{L^2(0,T; H^{M-1})}^2. \end{aligned} \quad (\text{C.1.5})$$

When $M = 0$, interpret the lower row in the inhomogeneous Bessel-potential space H^{-1} . The record $\mathcal{R}_{N,M}[u_0; T]$ retains these velocity norms, their normalized pressure rows, and the equations that identify all entries as coordinates of (u, p) . Smoothness pays every finite record once $T < T_*$ has been fixed.

Lemma C.1.1 (Rectangles on strict classical subintervals). *For each strict cutoff $T < T_*$ and each finite pair N, M ,*

$$\mathfrak{R}_{N,M}(u; T) < \infty. \quad (\text{C.1.6})$$

If $N' \leq N$, $M' \leq M$, and $T' \leq T$, forgetting the excess coordinates and restricting time produces the smaller record.

Proof. The closed interval $[0, T]$ stays strictly inside the classical lifespan. Persistence puts every fixed $\partial_t^n u$ continuously in every fixed H^q there, and compactness bounds that coordinate. Hence

$$\|\partial_t^n u\|_{L^2(0, T; H^q)}^2 \leq T \sup_{0 \leq t \leq T} \|\partial_t^n u(t)\|_{H^q}^2 < \infty.$$

Apply this inequality to the finitely many orders in (C.1.5). Restriction changes neither the solution nor its derivatives, which also proves compatibility. $\square$

Strict-cutoff finiteness is not terminal control. The scalar function $f(t) = (T_* - t)^{-1/2}$ already separates the two quantifiers:

$$f \in L^2(0, T) \quad \text{for every } T < T_*, \quad f \notin L^2(I_*).$$

This is not a fluid model; it only shows that every strict interval may be finite while the terminal integral diverges.

The preceding lemma permits exactly that growth. The producer theorem 3.4.1 supplies the missing fact: after a finite (N, M) is chosen, its bound is independent of the cutoff. For $N' \leq N$ and $M' \leq M$, define the terminal bonding map

$$\rho_{N', M'; I_*}^{N, M; I_*} : \mathcal{X}_{N, M}(I_*) \longrightarrow \mathcal{X}_{N', M'}(I_*)$$

by dropping the higher rows, embedding the retained velocity and pressure coordinates into the lower spaces, and recomputing the adjacent-row norm on I_* . Direct restriction and restriction in stages agree, while an unchanged pair gives the identity.

Proposition C.1.2 (Terminal realization of the rectangle estimate). *Assume Theorem 3.4.1. For every finite N, M , the datum-generated pressure-complete history satisfies*

$$\begin{aligned} \mathfrak{R}_{N,M}(u; I_*) &:= \sum_{n=0}^N \|\partial_t^n u\|_{L^2(I_*; H^{M+1})}^2 \\ &\quad + \sum_{n=0}^N \|\partial_t^{n+1} u\|_{L^2(I_*; H^{M-1})}^2 < \infty. \end{aligned} \tag{C.1.7}$$

The same coordinates therefore define a record on all of I_* :

$$\mathcal{R}_{N,M}[u_0] \in \mathcal{X}_{N,M}(I_*),$$

For every lower pair (N', M') , its terminal restriction satisfies

$$\rho_{N',M';I_*}^{N,M;I_*}(\mathcal{R}_{N,M}[u_0]) = \mathcal{R}_{N',M'}[u_0]. \tag{C.1.8}$$

All shared entries are the same distributional derivatives of u , and all pressure entries retain the normalization (C.1.2).

Proof. Fix the pair (N, M) and, for $0 \leq n \leq N$, set

$$f_n^+(t) := \|V_n(t)\|_{H^{M+1}}^2, \quad f_n^-(t) := \|V_{n+1}(t)\|_{H^{M-1}}^2.$$

The functions are nonnegative and $(0, T)$ increases to I_* . Monotone convergence gives, for either sign,

$$\int_{I_*} f_n^\sigma(t) dt = \lim_{T \uparrow T_*} \int_0^T f_n^\sigma(t) dt.$$

Each strict integral is one nonnegative term in the producer bound. Its limit is therefore finite, and

$$V_n \in L^2(I_*; H^{M+1}), \quad V_{n+1} \in L^2(I_*; H^{M-1}).$$

Summing the finitely many limits gives (C.1.7). The Riesz formula, Sobolev multiplication, and Cauchy–Schwarz in time also give

$$\|Q_n\|_{L^1(I_*; H^M)} \leq C_{M,n} \sum_{a=0}^n \|V_a\|_{L^2(I_*; H^{M+1})} \|V_{n-a}\|_{L^2(I_*; H^{M+1})} < \infty.$$

Thus the pressure rows belong to the terminal record. Dropping higher rows or lowering the spatial height changes no shared derivative or normalized pressure, proving (C.1.8). $\square$

For example, $(N, M) = (2, 3)$ retains V_0, V_1, V_2 in $L^2(I_*; H^4)$, V_1, V_2, V_3 in $L^2(I_*; H^2)$, and Q_0, Q_1, Q_2 in $L^1(I_*; H^3)$. Restriction to $(1, 2)$ drops V_3, Q_2 and reads the remaining coordinates in the corresponding lower spaces. Nothing is recomputed: each retained distribution is the same one. Compatibility concerns this identity; terminal boundedness was the separate quantifier paid by the producer theorem.

Each finite coordinate can now be read through its adjacent pair.

Proposition C.1.3 (Adjacent-row recovery). *For every finite n, m , the producer estimate gives*

$$V_n \in L^2(I_*; H^{m+1}), \quad \partial_t V_n = V_{n+1} \in L^2(I_*; H^{m-1}). \quad (\text{C.1.9})$$

The bound may depend on the chosen pair.

Proof. Choose $(N, M) = (n, m)$. The two terms indexed by n in (C.1.7) are precisely the displayed adjacent rows. They are actual derivatives of u , and restriction preserves that identity. $\square$

The strict and terminal records have always referred to the same derivatives of (u, p) . What has changed is the interval on which their norms are finite; (C.1.8) now makes the terminal family compatible under every decrease of either cutoff.

C.2 The Compatible Intersection

Every finite pair now has a terminal record. Enlarging either cutoff retains the lower record verbatim inside the larger one, because shared entries are the same distributional derivatives of u . Order finite pairs coordinatewise and assemble them with the bonding maps into

$$\mathcal{X}_\infty(I_*) := \varprojlim_{N, M} \mathcal{X}_{N, M}(I_*).$$

A family belongs to this inverse limit precisely when every finite coordinate is an admissible record and

$$\rho_{N', M'; I_*}^{N, M; I_*}(\mathbf{R}_{N, M}) = \mathbf{R}_{N', M'}$$

for every lower pair (N', M') .

Proposition C.2.1 (Compatible projective assembly). *The datum-generated rectangles define*

$$\mathcal{I}[u_0] := \left(\mathcal{R}_{N, M}[u_0] \right)_{N, M}.$$

This family lies in $\mathcal{X}_\infty(I_)$. Its velocity coordinate is the same distribution u throughout and obeys*

$$u \in \bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I_*; H^m(\mathbb{R}^3)). \quad (\text{C.2.1})$$

Coordinate projections commute with all bonding maps.

Proof. Given finite r, m , choose $N \geq r$ and $M \geq m$. That rectangle contains

$$\partial_t^n u \in L^2(I_*; H^{M+1}) \hookrightarrow L^2(I_*; H^m) \quad (0 \leq n \leq r).$$

Because these are the distributional time derivatives of one velocity, $u \in H^r(I_*; H^m)$. The pair was arbitrary, proving (C.2.1). Compatibility makes each lower record the literal projection of every larger one, so no representatives have to be selected or reconciled. $\square$

Project now to the zeroth temporal row while allowing every finite spatial height:

$$\mathcal{S}_0(I_*) := \bigcap_{m=0}^{\infty} L^2(I_*; H^m(\mathbb{R}^3)). \quad (\text{C.2.2})$$

Each $\mathcal{R}_{0,m}[u_0]$ supplies one coordinate, and compatibility identifies them all with the same u . Hence

$$\pi_0(\mathcal{I}[u_0]) = u \in \mathcal{S}_0(I_*).$$

Every finite spatial energy used in the derivative ascent is a coordinate of this row. For $s \geq 0$, set

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2. \quad (\text{C.2.3})$$

Choose one integer $m \geq s$. Its coordinate gives

$$\int_0^{T_*} E_s(t) dt \leq C_{s,m} \|u\|_{L^2(I_*; H^m)}^2 < \infty. \quad (\text{C.2.4})$$

Each finite energy is therefore integrable on I_* , with its own bound and the same underlying velocity.

C.3 The Critical Height at the Terminal Face

The terminal rectangles provide adjacent spatial rows. Their first task is to turn two time-integrated norms into one strong value at the terminal face.

Lemma C.3.1 (Adjacent Hilbert-scale trace). *Let $s \in \mathbb{R}$ and suppose*

$$v \in L^2(I_*; H^{s+1}), \quad v_t \in L^2(I_*; H^{s-1}).$$

Then v has one representative in $C([0, T_]; H^s)$, and for $0 \leq a \leq b \leq T_*$,*

$$\left| \|v(b)\|_{H^s}^2 - \|v(a)\|_{H^s}^2 \right| \leq 2 \|v_t\|_{L^2(a,b; H^{s-1})} \|v\|_{L^2(a,b; H^{s+1})}. \quad (\text{C.3.1})$$

Proof. Let $A = (1 - \Delta)^{1/2}$. Time mollification and frequency truncation make the following identity legitimate for smooth approximants:

$$\|v(b)\|_{H^s}^2 - \|v(a)\|_{H^s}^2 = 2 \operatorname{Re} \int_a^b \langle A^{s-1} v_t(t), A^{s+1} v(t) \rangle_{L^2} dt. \quad (\text{C.3.2})$$

Cauchy–Schwarz gives (C.3.1). The two assumed L^2 rows pass the right-hand side to the limit. The limit identity makes the H^s norm continuous, while the approximation also gives weak H^s continuity; together they imply strong continuity. Any two continuous representatives agree everywhere because they agree almost everywhere. $\square$

The critical readout needs only $\mathcal{R}_{0,2}[u_0]$. Its integer rows embed into the fractional adjacent pair

$$u \in L^2(I_*; H^{5/2}), \quad u_t \in L^2(I_*; H^{1/2}). \quad (\text{C.3.3})$$

No fractional rectangle has been assumed; these are continuous embeddings of the integer record. The trace lemma at $s = 3/2$ now gives

$$u \in C([0, T_*]; H^{3/2}). \quad (\text{C.3.4})$$

Let $H = E_{1/2}$. To read the homogeneous critical energies from the inhomogeneous trace, mollify in time and truncate frequencies to $\varepsilon \leq |\xi| \leq R$. Differentiation and Plancherel then give

$$\frac{1}{2} \left\| \Lambda^{3/2} u(b) \right\|_{L^2}^2 - \frac{1}{2} \left\| \Lambda^{3/2} u(a) \right\|_{L^2}^2 = \operatorname{Re} \int_a^b \left\langle \Lambda^{1/2} u_t(t), \Lambda^{5/2} u(t) \right\rangle_{L^2} dt.$$

Cauchy–Schwarz passes the integral to the limit through the two critical rows; the continuous $H^{3/2}$ trace passes the endpoint terms. Removing all cutoffs yields, for $0 \leq a \leq b \leq T_*$,

$$E_{3/2}(b) - E_{3/2}(a) = \operatorname{Re} \int_a^b \left\langle \Lambda^{1/2} u_t(t), \Lambda^{5/2} u(t) \right\rangle_{L^2} dt.$$

The adjacent pair makes the derivative integrable, and the complete spatial row makes $E_{3/2}$ itself integrable. Hence

$$E_{3/2} \in W^{1,1}(I_*) \cap L^1(I_*) \subset L^\infty(I_*). \quad (\text{C.3.5})$$

The integer rows also put the derivative of the critical height in L^1 :

$$H'(t) = \left\langle \Lambda^{1/2} u(t), \Lambda^{1/2} u_t(t) \right\rangle_{L^2} \in L^1(I_*). \quad (\text{C.3.6})$$

Keep the pressure-explicit production current as

$$\mathcal{P}_{1/2} := -\operatorname{Re} \left\langle \Lambda^{1/2} u, \Lambda^{1/2} \left((u \cdot \nabla) u + \nabla p \right) \right\rangle_{L^2}.$$

Pairing the full equation with Λu gives

$$H' = -2\nu E_{3/2} + \mathcal{P}_{1/2}.$$

The two integrable rows consequently imply

$$H \in W^{1,1}(I_*), \quad E_{3/2} \in L^\infty(I_*) \cap L^1(I_*), \quad \mathcal{P}_{1/2} \in L^1(I_*).$$

The height and $E_{3/2}$ therefore reach the terminal face continuously; the production current is only integrable, and no endpoint value is claimed for it. This entire critical trace comes from $\mathcal{R}_{0,2}[u_0]$. Higher rectangles will repeat the endpoint argument at every finite Sobolev height.

C.4 One Endpoint for the Whole Tower

A restart needs an actual datum at T_* , not a separate endpoint in each rectangle. The complete spatial row contains every finite spatial height; the equation supplies enough time integrability to make the same velocity converge at all of them.

Lemma C.4.1 (The complete spatial row reaches every endpoint topology). *Let (u, p) be the fixed normalized history on I_* . If $u \in \mathcal{S}_0(I_*)$, then for every $m \in \mathbb{N}_0$,*

$$u_t \in L^1(I_*; H^m), \quad u \in W^{1,1}(I_*; H^m) \hookrightarrow C([0, T_*]; H^m). \quad (\text{C.4.1})$$

Proof. The pressure-complete transport response can be written without separating pressure:

$$(u \cdot \nabla)u + \nabla p = \mathbb{P} \nabla \cdot (u \otimes u).$$

Under the fixed normalization, this is exactly transport plus ∇p . Sobolev multiplication and boundedness of the Leray and Riesz multipliers give

$$\|\mathbb{P} \nabla \cdot (u \otimes u)\|_{H^m} \leq C_m \|u\|_{H^{m+2}}^2.$$

The complete row, together with $T_* < \infty$, therefore yields

$$\begin{aligned} \|\mathbb{P} \nabla \cdot (u \otimes u)\|_{L^1(I_*; H^m)} &\leq C_m \|u\|_{L^2(I_*; H^{m+2})}^2 < \infty, \\ \|\Delta u\|_{L^1(I_*; H^m)} &\leq T_*^{1/2} \|u\|_{L^2(I_*; H^{m+2})} < \infty. \end{aligned}$$

The momentum equation puts u_t in $L^1(I_*; H^m)$. In particular, for $0 \leq s < t < T_*$,

$$\|u(t) - u(s)\|_{H^m} \leq \nu(t-s)^{1/2} \|u\|_{L^2((s,t); H^{m+2})} + C_m \|u\|_{L^2((s,t); H^{m+2})}^2.$$

Both terms vanish when $s, t \uparrow T_*$, making the same $u(t)$ Cauchy in H^m . Since the finite interval also embeds $L_t^2 H_x^m$ into $L_t^1 H_x^m$, $u \in W^{1,1}(I_*; H^m)$; the Bochner fundamental theorem of calculus supplies the continuous representative and its endpoint. $\square$

Let $u_*^{(m)}$ be the endpoint found in H^m . If $k > m$, both the H^k trace embedded into H^m and the native H^m trace represent the same u almost everywhere. Continuity makes them identical through the closed interval. Their terminal values are therefore one element

$$u_* := u(T_*) \in \bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3). \quad (\text{C.4.2})$$

No uniformity in m is used. Divergence passes continuously to the limit, so u_* is solenoidal, and the strong L^2 trace preserves finite kinetic energy.

Apply the same adjacent trace to every temporal coordinate:

$$\partial_t^N u \in C([0, T_*]; H^m) \quad (N, m \in \mathbb{N}_0).$$

For fixed N , spatial compatibility identifies all these traces as

$$V_N^* := \lim_{t \uparrow T_*} \partial_t^N u(t) \in H^\infty(\mathbb{R}^3). \quad (\text{C.4.3})$$

It remains to show that this trace family is the actual Navier–Stokes jet generated by u_* .

Proposition C.4.2 (Pressure-complete endpoint jet). *The endpoint u_* determines one sequence $(V_N^*, Q_N^*)_{N \geq 0} \subset H^\infty \times H^\infty$ by*

$$\begin{aligned} V_0^* &= u_*, \\ Q_N^* &= R_i R_j \sum_{a=0}^N \binom{N}{a} V_{a,i}^* V_{N-a,j}^*, \end{aligned} \tag{C.4.4}$$

$$V_{N+1}^* = \nu \Delta V_N^* - \sum_{a=0}^N \binom{N}{a} (V_a^* \cdot \nabla) V_{N-a}^* - \nabla Q_N^*. \tag{C.4.5}$$

This sequence is exactly the family of left traces of the original normalized velocity and pressure history.

Proof. The space $H^\infty = \bigcap_m H^m$ is stable under multiplication and spatial differentiation, and Riesz transforms preserve every one of its Sobolev coordinates. The two recurrences therefore determine a unique next pair at each finite order.

Fix a finite target height m and take the trace convergence in a Sobolev space high enough for every product and derivative in the recurrence. All operations are continuous between these finite spaces, so the limit of (C.1.3)–(C.1.4) is the displayed endpoint recurrence. Starting from $V_0^* = u_*$, induction in N identifies the recursively generated entries with the left traces. Since m was arbitrary, the identity holds in H^∞ , and spatial differentiation supplies every mixed endpoint coordinate. $\square$

No new pressure constant appears at the terminal face. The decaying Riesz formula fixes every $Q_N(t)$ on the left, and the limit preserves that normalization. Thus inserting u_* into the normalized recurrence produces exactly the left mixed jet.

C.5 Restarting the Same History

Use the endpoint itself as the restart datum. The local theorem gives a right-hand solution on one interval for all finite Sobolev heights. Its jet and the left trace jet both begin from u_* and obey the same normalized recurrence; the join is therefore smooth, and uniqueness makes the two pieces one history.

Lemma C.5.1 (Smooth local restart and uniqueness). *Let $t_0 \geq 0$ and $a \in H_\sigma^\infty(\mathbb{R}^3)$. There are $\delta > 0$ and a unique smooth normalized pair (w, q) on $[t_0, t_0 + \delta] \times \mathbb{R}^3$, with $w(t_0) = a$ and*

$$q = R_i R_j (w_i w_j).$$

All finite spatial Sobolev orders persist on this same interval. The temporal jet at t_0 is generated by (C.4.4)–(C.4.5).

Proof. Translate time and apply Proposition A.1.1 to a . Its lifespan depends only on ν and $\|a\|_{H^3}$, while persistence carries every finite height of a on that interval. The normalized pair

is unique. Differentiating at the initial face and using the Riesz pressure formula gives the endpoint recurrences. $\square$

Theorem C.5.2 (Global continuation from whole-terminal rectangles). *Let u_0 and all its spatial derivatives be rapidly decaying, with $\nabla \cdot u_0 = 0$, and fix $\nu > 0$. Let (u, p) be its maximal normalized classical history. If the whole-terminal compatible rectangles of Proposition C.1.2 exist, then $T_* = \infty$ and (u, p) is smooth on every finite interval.*

Proof. Suppose instead that $T_* < \infty$. The hypothesis supplies every terminal rectangle with compatible overlaps. Their intersection produces the common endpoint $u_* \in H_\sigma^\infty$ and its actual normalized jet. Restart the local theorem at T_* from u_* , obtaining (w, q) on $[T_*, T_* + \delta]$, and set

$$(\tilde{u}, \tilde{p})(t) := \begin{cases} (u, p)(t), & 0 \leq t < T_*, \\ (w, q)(t), & T_* \leq t \leq T_* + \delta. \end{cases}$$

The two velocities meet at u_* . Every one-sided pressure and velocity derivative also agrees because both jets are generated from that datum by the same normalized recurrences. The piecewise pair is smooth across T_* and solves the same fixed-viscosity equation beyond it. Strong uniqueness makes it the continuation of the original datum-generated history, contradicting maximality. Therefore

$$T_* = \infty.$$

On each finite interval, every finite temporal and spatial coordinate remains in the compatible intersection. Sobolev embedding then gives

$$u, p \in C^\infty([0, \infty) \times \mathbb{R}^3).$$

$\square$

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